

第七讲：图像分割



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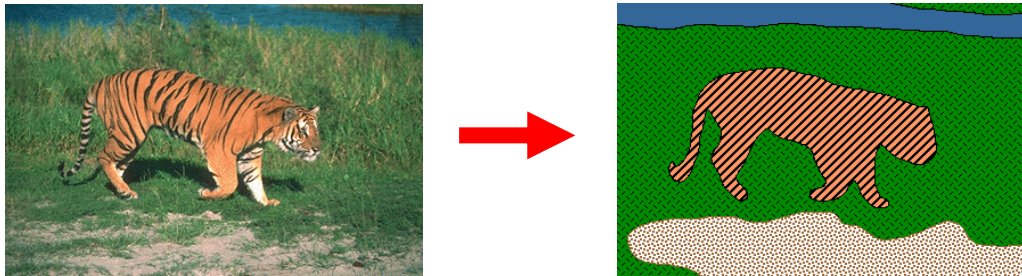
2026-3-24



What is segmentation 分割的概念

- The process of partitioning an image into **multiple regions**
 - Sets of pixels (different objects or super pixels)
- Goal is to **simplify the representation** of an image
 - Makes it more meaningful and easier to analyze
- Helpful for detecting objects and boundaries

将图像中不同类型的区域划分出来

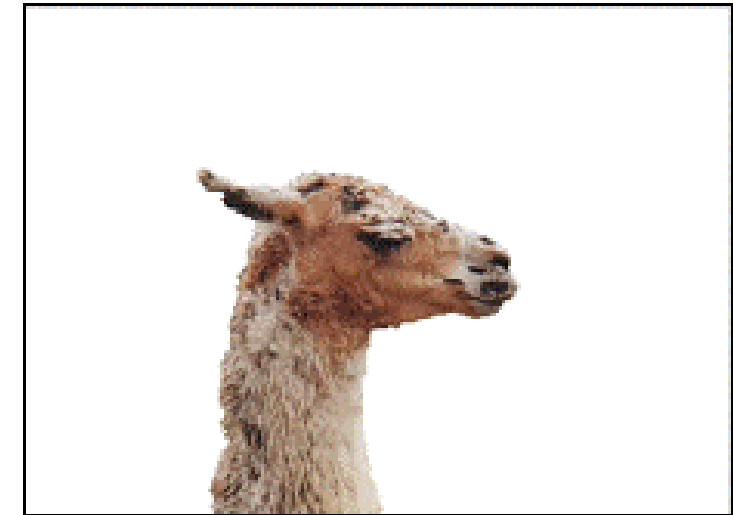
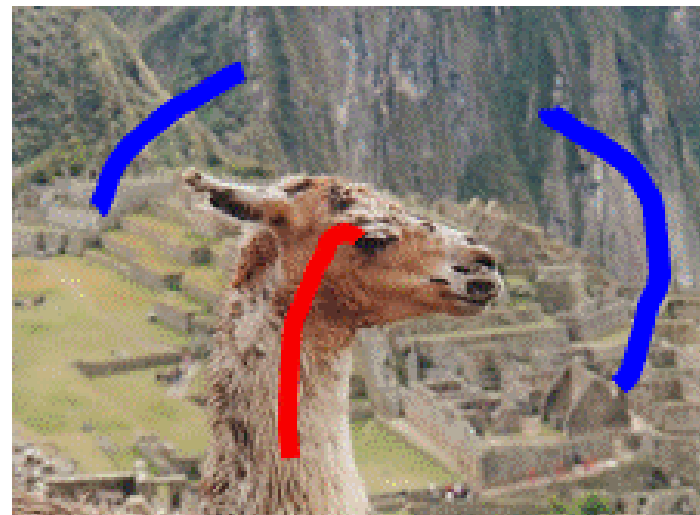


可看做像素级的分类，或寻找区域轮廓的过程



What is segmentation 分割的概念

- Special case: **binary segmentation 二值分割**
- Obtain the foreground and background region



Outline



1, Intensity-based Segmentation

Thresholding 阈值法

Watershed 分水岭算法

2, Clustering-based Segmentation

3, Active contours

Snake 蛇形模型

Level set 水平集

Thresholding-based Segmentation 基于阈值的分割



- Consider pixel value only 根据灰度阈值将图像分为前景和背景
 - Simple and easy to implement
 - Shape of histogram is often used for automation

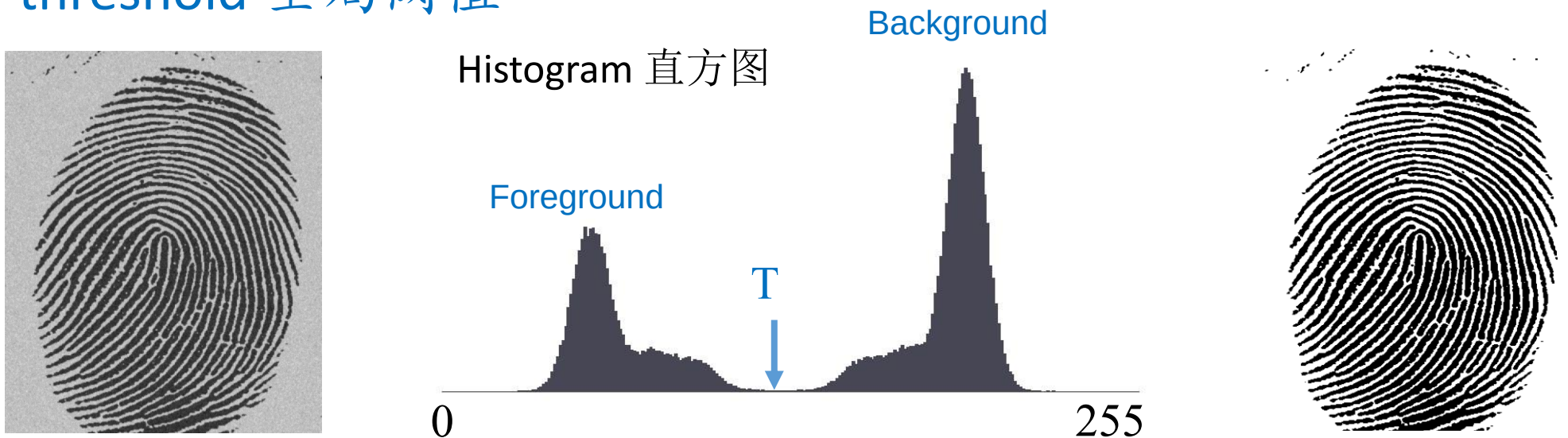


A **threshold T** , a gray level intensity, classifies every pixel as belonging to objects (foreground) or background.

- Global threshold
- Local threshold
- Optimal threshold

Thresholding-based Segmentation 基于阈值的分割

- Global threshold 全局阈值



We chose a threshold T midway between the two gray value distributions.

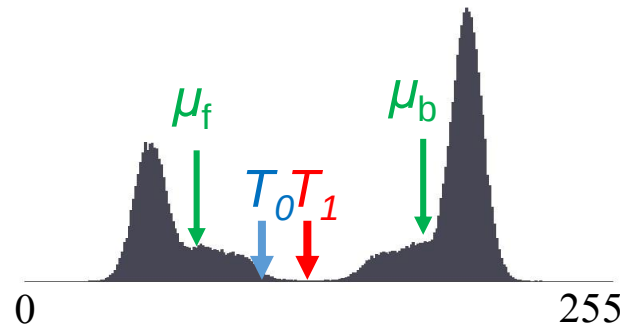
Thresholding-based Segmentation 基于阈值的分割



- How to find a global threshold

寻找全局阈值的算法过程

1. Choose initial threshold T_0
2. Define $f(x,y) > T_0$ as background and $f(x,y) < T_0$ as foreground
3. Calculate mean for background μ_b and foreground μ_f
4. Set next threshold $T_1 = (\mu_b + \mu_f)/2$
5. Repeat 2.-4. until stopping criteria, $T_i = T_{i-1}$, is fulfilled

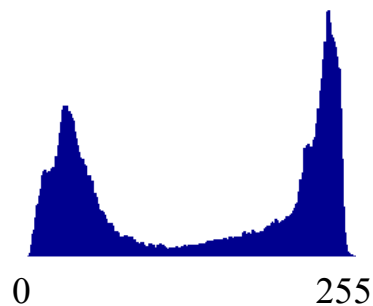
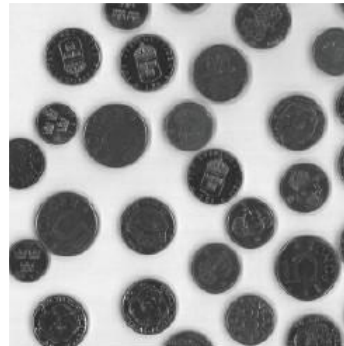


Thresholding-based Segmentation 基于阈值的分割

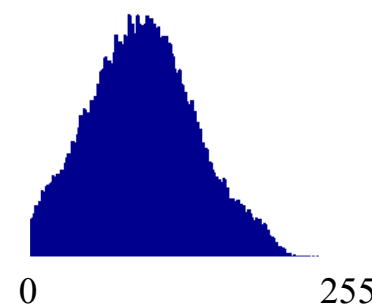


- When does global intensity-based thresholding work and not work?

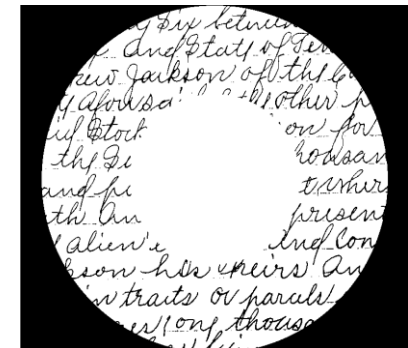
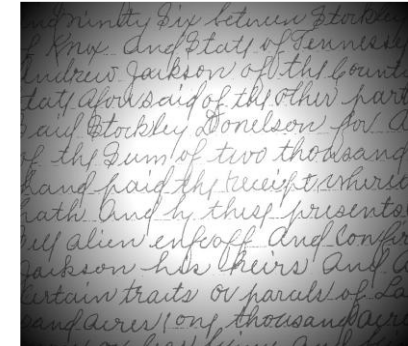
Foreground and background intensity are quite different



Foreground and background have similar intensity values



Non-uniform intensity

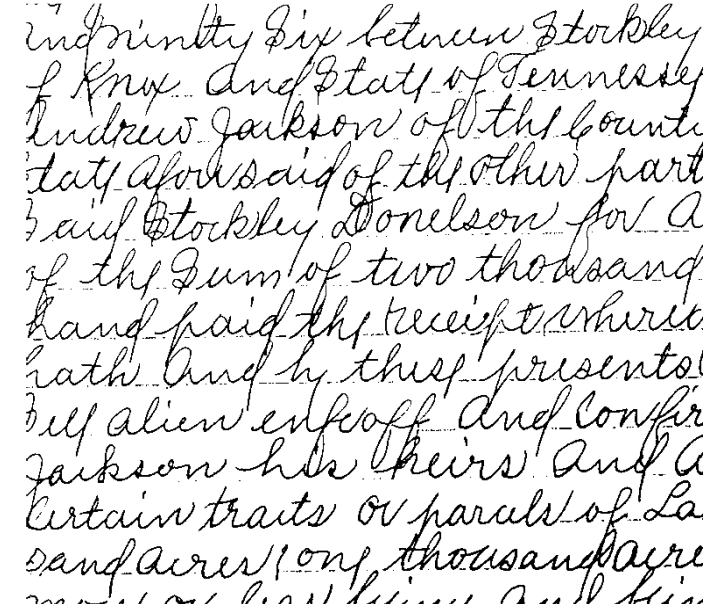
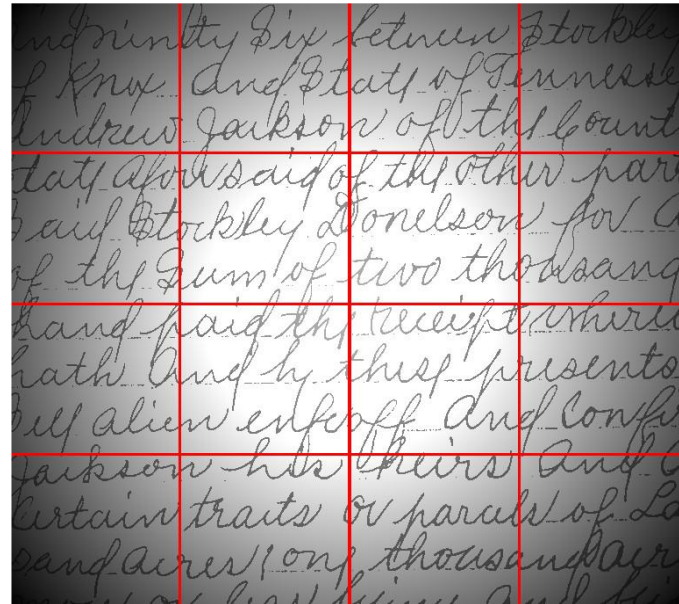
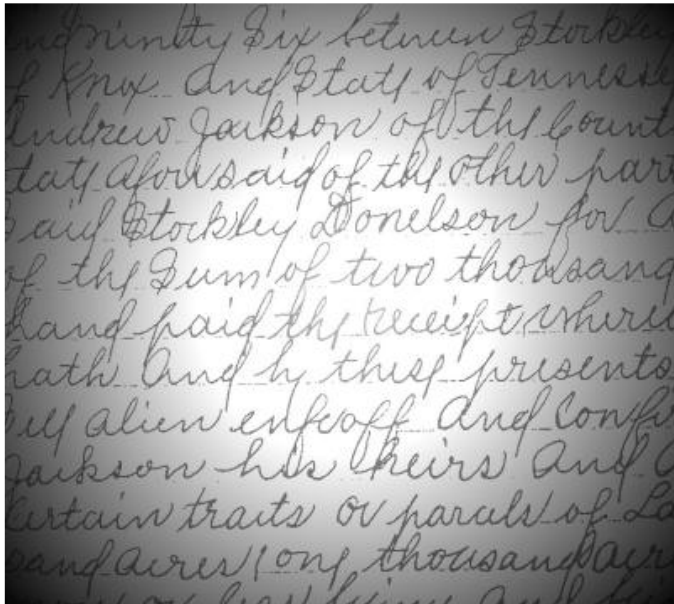


全局阈值的局限性

Thresholding-based Segmentation 基于阈值的分割



- Local thresholding 局部阈值



Example method

每个局部范围内的灰度比较均匀

Subdivide image into non-overlapping rectangles. These rectangles are chosen small enough so that the **illumination of each is approximately uniform**. Then determine a global threshold for each sub-image.

Thresholding-based Segmentation 基于阈值的分割



- Optimal thresholding: Otsu 大津算法——求最佳阈值
 - Determine the threshold by minimizing intra-class intensity variance
 - Or equivalently, maximizing inter-class variance

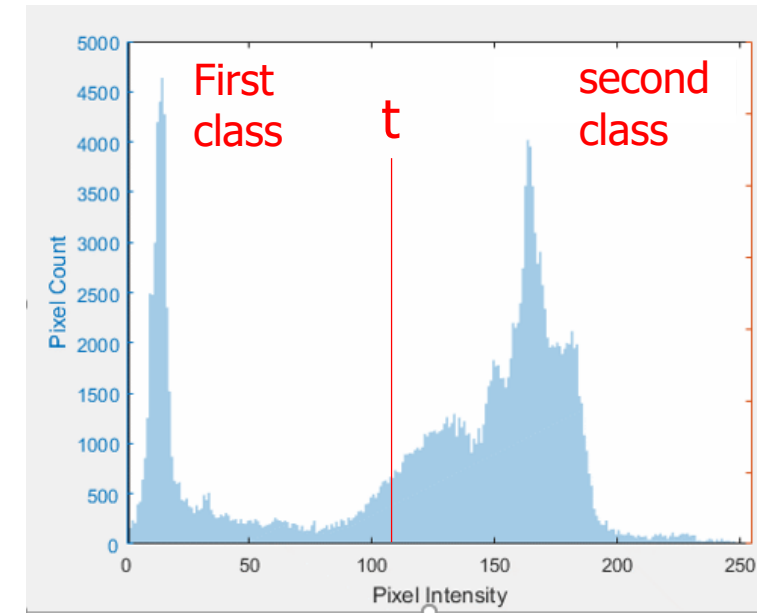
$$\sigma_w^2(t) = \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t)$$

variance of first class variance of second class

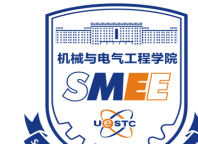
Probability of first class Probability of second class

$$\omega_0(t) = \sum_{i=0}^{t-1} p(i)$$

$$\omega_1(t) = \sum_{i=t}^{L-1} p(i)$$



Thresholding-based Segmentation 基于阈值的分割



• Optimal thresholding: Otsu

$$\sigma_w^2(t) = \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t)$$



类内方差最小化 等效于 类间方差最大化



$$\begin{aligned}\sigma_b^2(t) &= \sigma^2 - \sigma_w^2(t) = \omega_0(t)(\mu_0 - \mu_T)^2 + \omega_1(t)(\mu_1 - \mu_T)^2 \\ &= \omega_0(t)\omega_1(t)[\mu_0(t) - \mu_1(t)]^2\end{aligned}$$

$$\mu_0(t) = \frac{\sum_{i=0}^{t-1} ip(i)}{\omega_0(t)}$$

$$\mu_1(t) = \frac{\sum_{i=t}^{L-1} ip(i)}{\omega_1(t)}$$

$$\mu_T = \sum_{i=0}^{L-1} ip(i)$$

$$\omega_0\mu_0 + \omega_1\mu_1 = \mu_T$$

$$\omega_0 + \omega_1 = 1$$

$$\begin{aligned}\sigma^2 - \sigma_w^2 &= \sigma^2 - \omega_0\sigma_0^2 - \omega_1\sigma_1^2 \\ &= \sum_{i=0}^{L-1} p_i(i - \mu_T)^2 - \omega_0 \frac{\sum_{i=0}^t p_i(i - \mu_0)^2}{\omega_0} - \omega_1 \frac{\sum_{i=t+1}^{L-1} p_i(i - \mu_1)^2}{\omega_1} \\ &= \underbrace{\sum_{i=0}^t p_i(i - \mu_T)^2 - \sum_{i=0}^t p_i(i - \mu_0)^2}_{\sum_{i=0}^t p_i((i - \mu_T)^2 - (i - \mu_0)^2)} + \underbrace{\sum_{i=t+1}^{L-1} p_i(i - \mu_T)^2 - \sum_{i=t+1}^{L-1} p_i(i - \mu_1)^2}_{\sum_{i=t+1}^{L-1} p_i((i - \mu_T)^2 - (i - \mu_1)^2)} \\ &= \sum_{i=0}^t p_i(2i - \mu_T - \mu_0)(\mu_0 - \mu_T) \\ &= \sum_{i=0}^t (2ip_i - p_i(\mu_0 + \mu_T))(\mu_0 - \mu_T) \\ &= (\mu_0 - \mu_T) \left[2 \sum_{i=0}^t ip_i - \sum_{i=0}^t p_i(\mu_0 + \mu_T) \right] \\ &= (\mu_0 - \mu_T) [2\omega_0\mu_0 - (\mu_0 + \mu_T)\omega_0] \\ &= \omega_0(\mu_0 - \mu_T)^2 + \omega_1(\mu_1 - \mu_T)^2\end{aligned}$$

$$\text{故} \lambda \quad \mu_T = \omega_0\mu_0 + \omega_1\mu_1$$

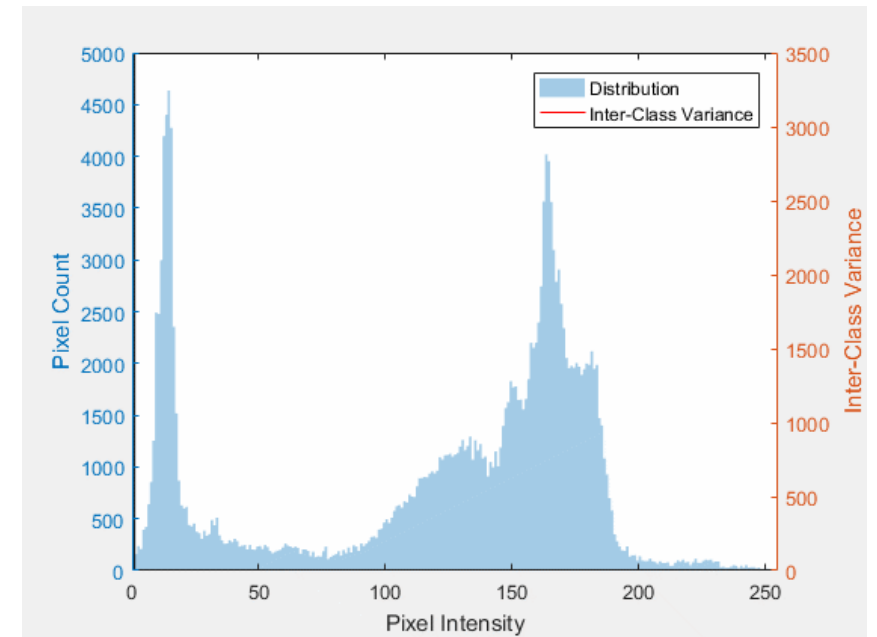
$$\begin{aligned}\text{上式} &= \omega_0[\mu_0 - \omega_0\mu_0 - \omega_1\mu_1]^2 + \omega_1[\mu_1 - \omega_0\mu_0 - \omega_1\mu_1]^2 \\ &= \omega_0[\omega_1\mu_0 - \omega_1\mu_1]^2 + \omega_1[\omega_0\mu_1 - \omega_0\mu_0]^2 \\ &= \omega_0\omega_1^2(\mu_0 - \mu_1)^2 + \omega_1\omega_0^2(\mu_0 - \mu_1)^2 \\ &= \omega_0\omega_1(\mu_0 - \mu_1)^2(\omega_1 + \omega_0) \\ &= \omega_0\omega_1(\mu_0 - \mu_1)^2\end{aligned}$$

Thresholding-based Segmentation 基于阈值的分割

- Optimal thresholding: Otsu

$$\sigma_w^2(t) = \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t)$$

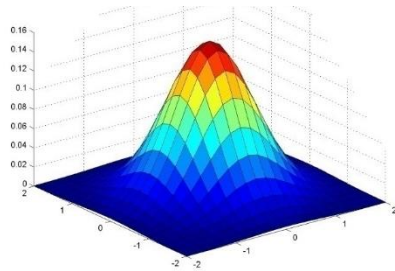
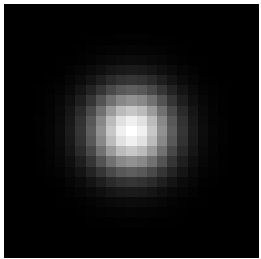
$$\begin{aligned}\sigma_b^2(t) &= \sigma^2 - \sigma_w^2(t) = \omega_0(t)(\mu_0 - \mu_T)^2 + \omega_1(t)(\mu_1 - \mu_T)^2 \\ &= \omega_0(t)\omega_1(t)[\mu_0(t) - \mu_1(t)]^2\end{aligned}$$



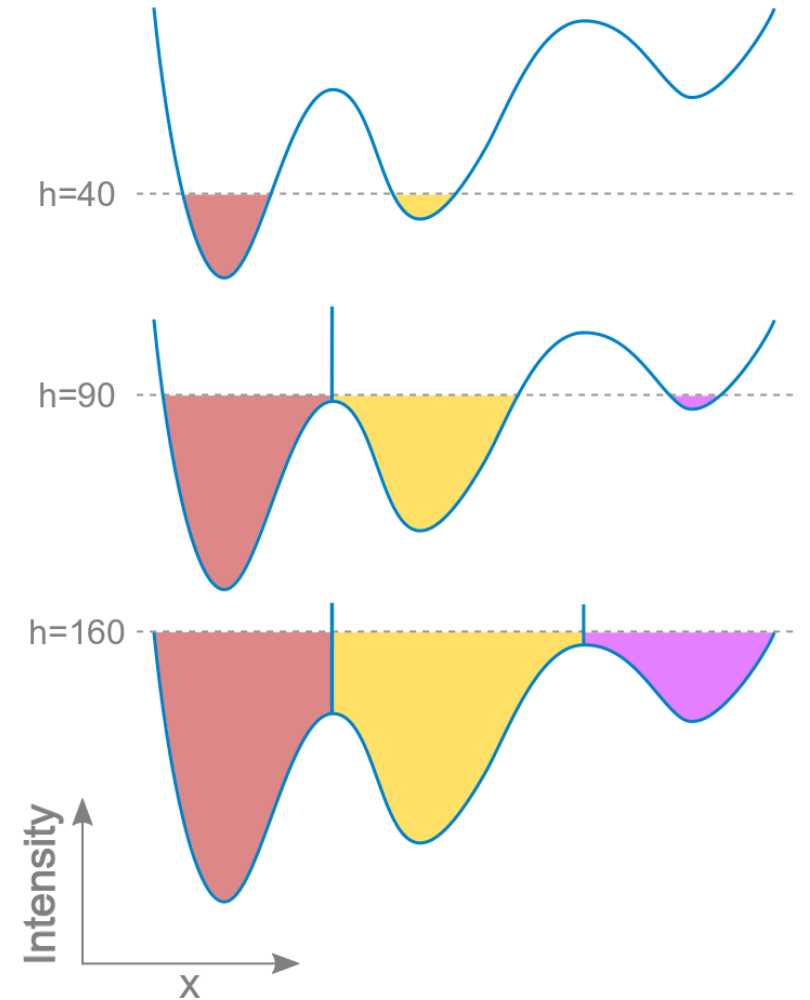
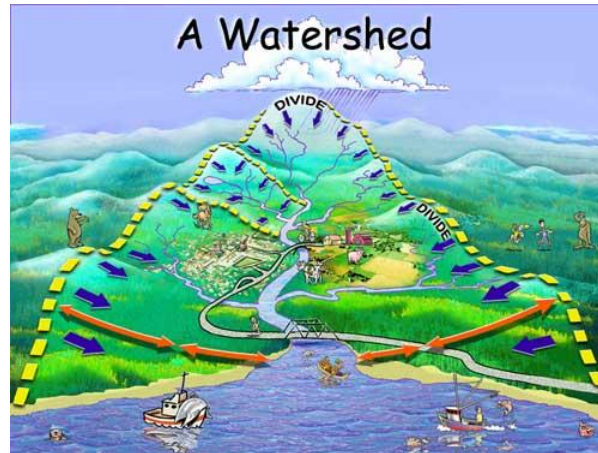
Watershed 分水岭算法

把灰度图像看做一个地势图

- Think the gray level image as a landscape
 - Let water rise from the bottom of each valley
 - As soon as the water from two valleys meet, build a dam, or a watershed
 - These watersheds will define the borders between different regions in the image

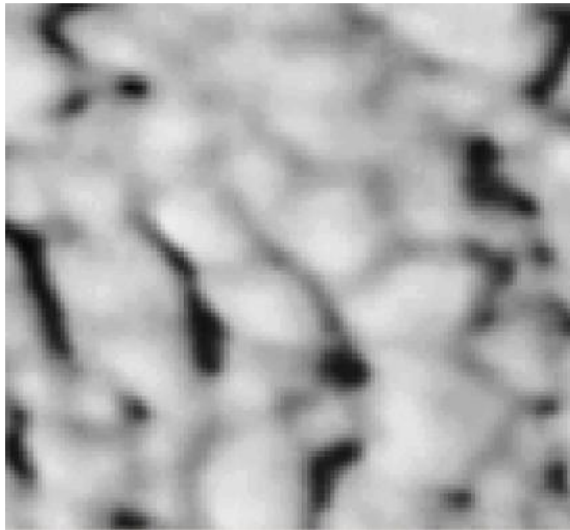


亮度值当做高度值

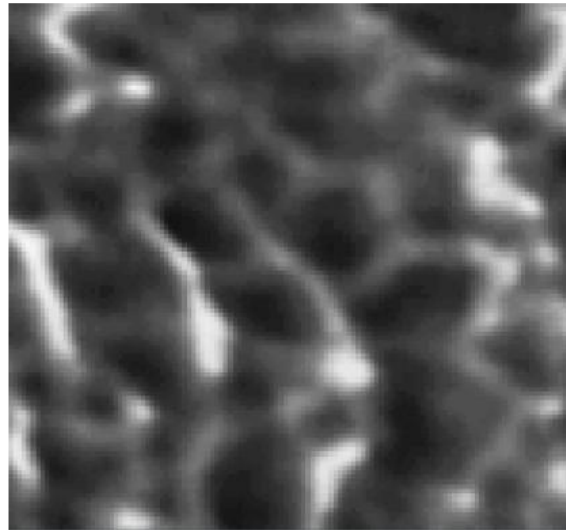


Watershed 分水岭算法

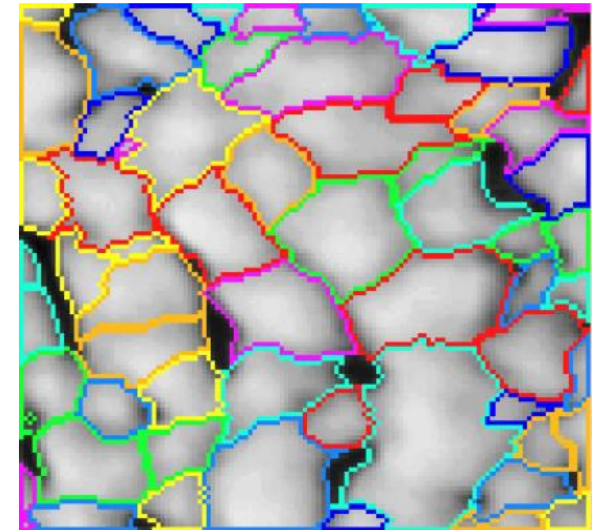
- Example of watershed directly applied on gray level image



Original image



Inverted image
(starting points
are in valleys)



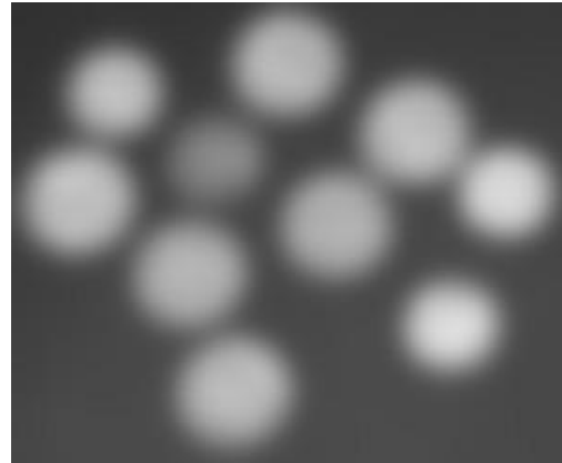
Segmentation result

Watershed 分水岭算法

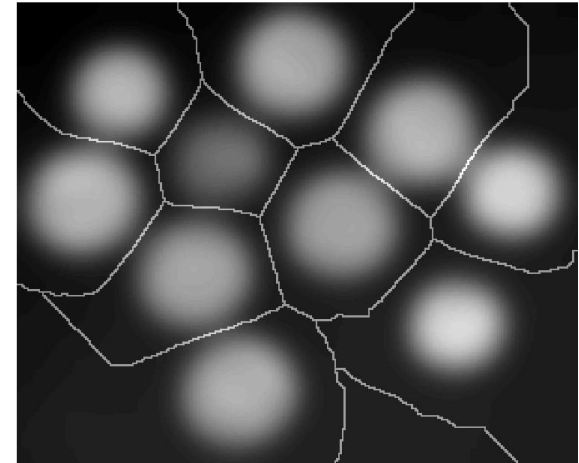
- Example of watershed directly applied on gray level image



Original image



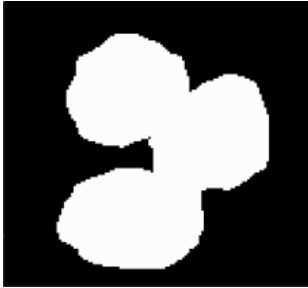
Blurred image



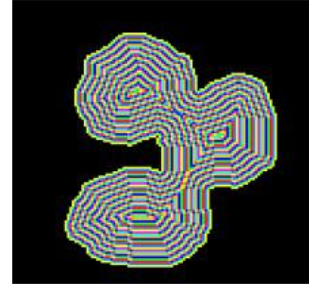
Segmentation result

Watershed 分水岭算法

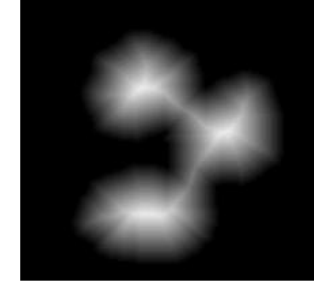
- Example of watershed on distance transformed image



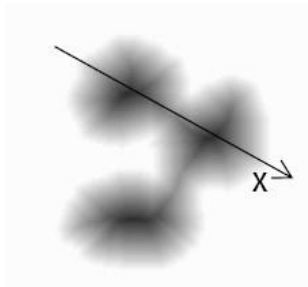
Original image



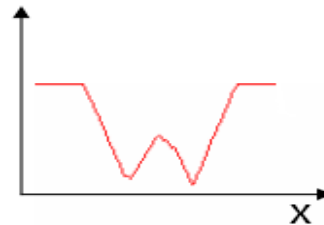
Distance transform



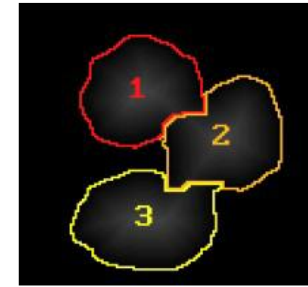
DT as intensity



DT inverse



Intensity landscape



Segmentation result

Outline



1, Intensity-based Segmentation

- Thresholding

- Watershed

2, Clustering-based Segmentation

3, Active contours

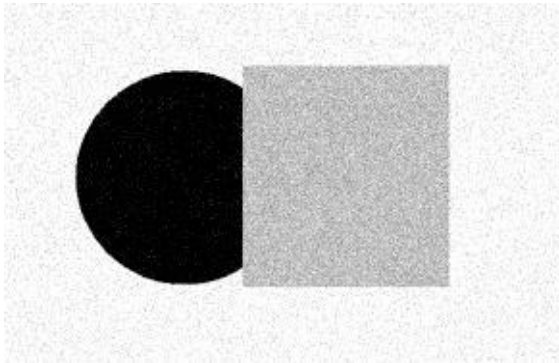
- Snake

- Level set

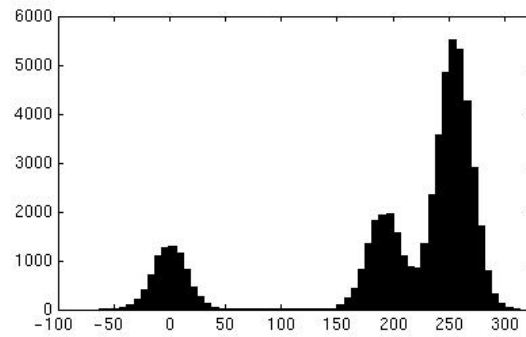
Clustering-based Segmentation 基于聚类的分割



- Clustering split pixels in to groups based on their **similarity**



image



histogram



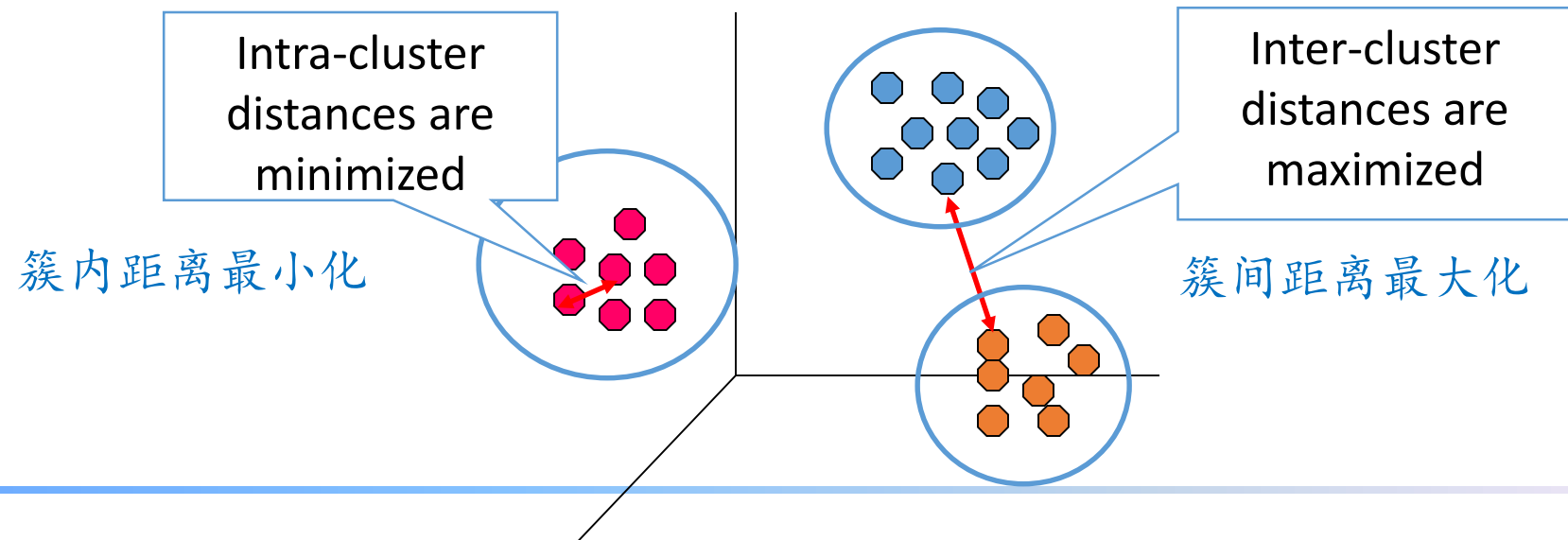
How do we determine the 3 main intensities that define our groups?

The foreground have similar colors, that are different from the background

Clustering-based Segmentation 基于聚类的分割



- What is clustering? 聚类的概念
- Unsupervised data analysis
- Finding groups of objects such that:
 - Objects in a group will be similar (or related) to one another
 - Objects in different groups are different (or unrelated)



Clustering-based Segmentation 基于聚类的分割



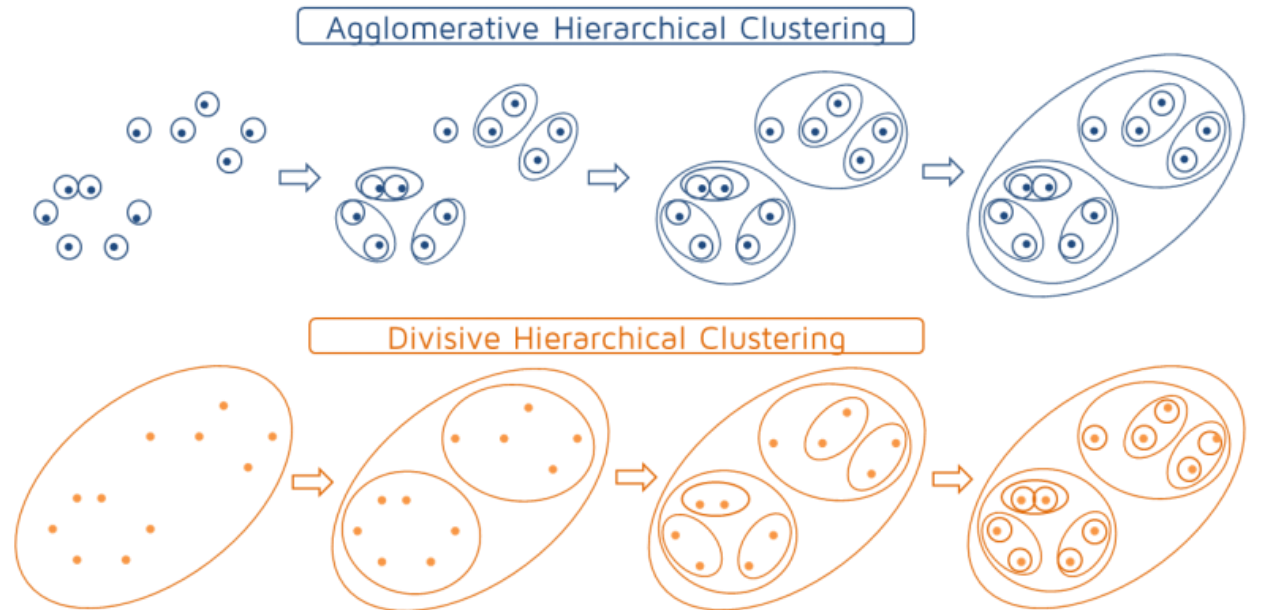
- Cluster pixels together pixels that are similar

- Agglomerative clustering 聚合聚类**

- Start from a single pixel
- Attach pixel to the closest cluster
- repeat

- Divisive clustering 分裂聚类**

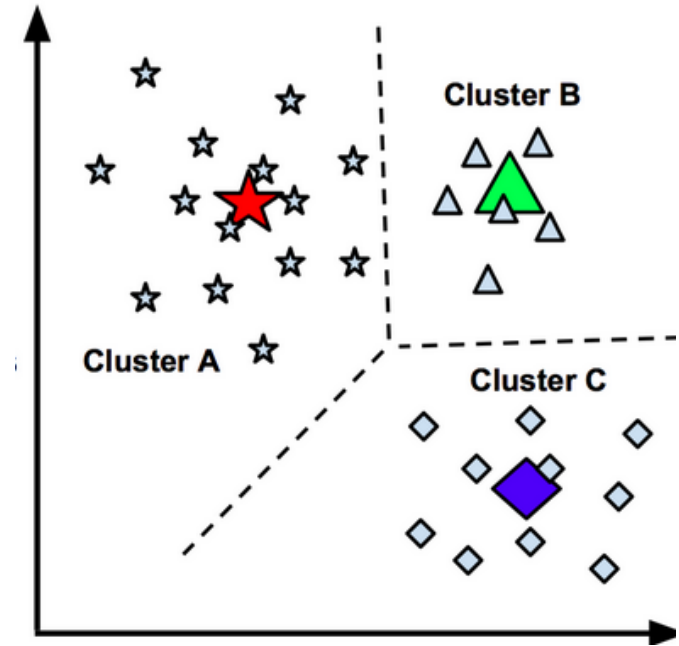
- Start from the entire image
- Split cluster along best boundary
- repeat



Clustering-based Segmentation 基于聚类的分割



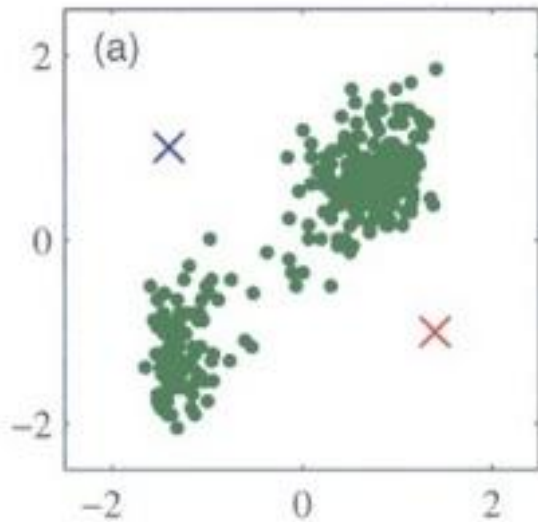
- K-means clustering K-均值聚类
- Partitional and center-based approach
 - Number of clusters K must be specified
 - Each cluster is associated with a centroid
 - Each point is assigned to the cluster with the closest centroid
- The basic algorithm is simple
 - 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change



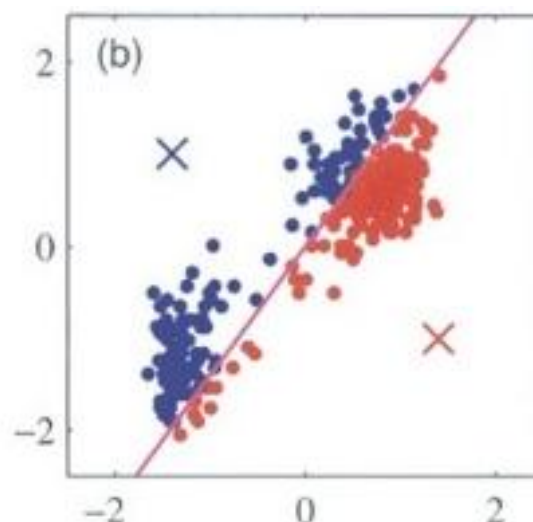
Clustering-based Segmentation 基于聚类的分割



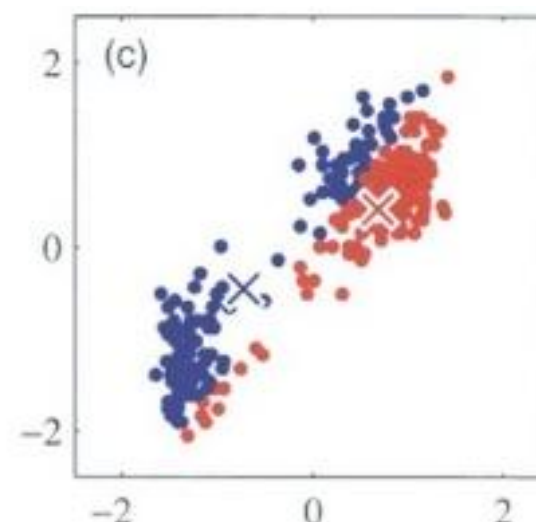
- K-means clustering
 - Update cluster centroids and membership alternatively



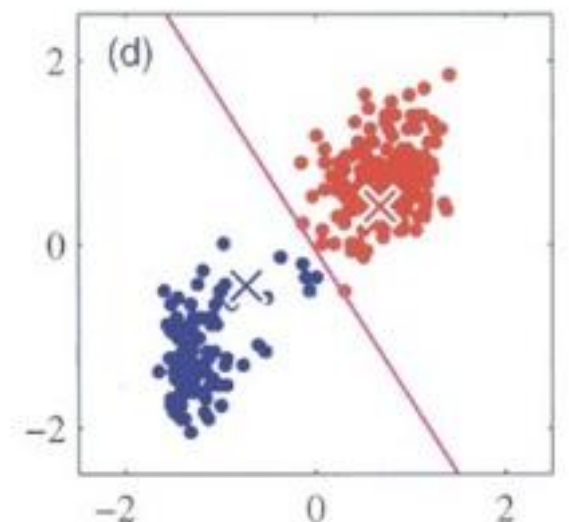
Initial centroids



Initial clusters



Updated centroids



Updated clusters

Image segmentation by K-means

- Select a value of K
- Select a feature vector for every pixel
 - e.g., color, texture, position, or combination
- Define a similarity measure between feature vectors
 - Usually Euclidean Distance
- Apply K-Means Algorithm
- Post-process (“clean up”) results

Clustering-based Segmentation 基于聚类的分割



Results of segmentation by K-means



image



Clusters on **intensity**



Clusters on **color**

SLIC Super-pixel 超像素



Outline



1, Intensity-based Segmentation

- Thresholding

- Watershed

2, Clustering-based Segmentation

3, Active contours

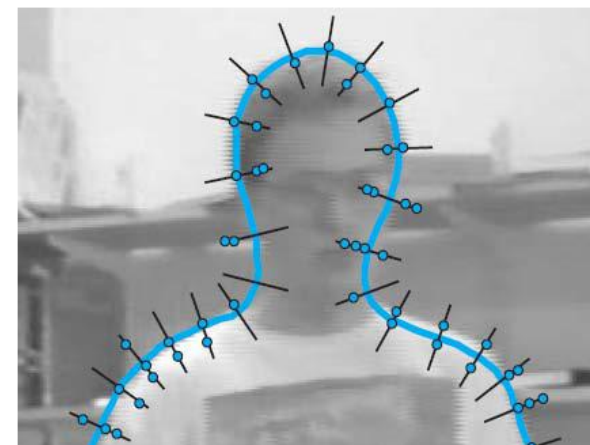
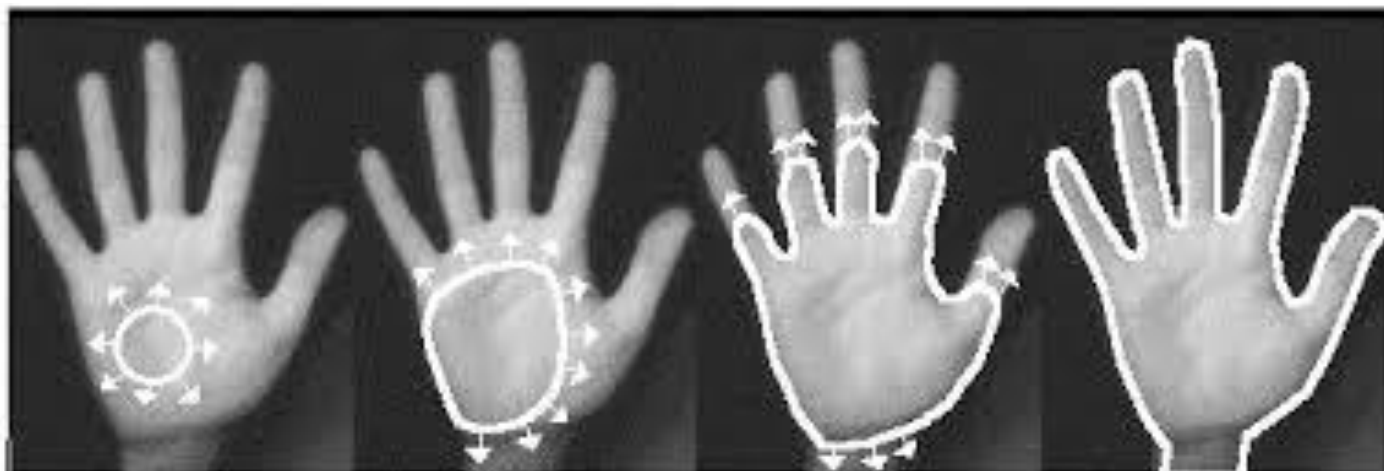
- Snake

- Level set

Active contours 活动轮廓

Treat the segmentation as a deformation of a curve

- Define a **energy** for the contour and minimize the energy
- Can obtain smooth segmentation contours
- Need multiple steps for curve deformation
- Typical methods: **snake** and **level set**

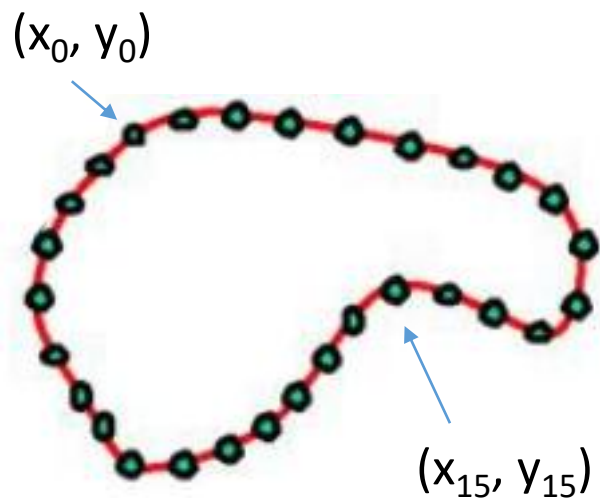


Snake model 蛇形模型

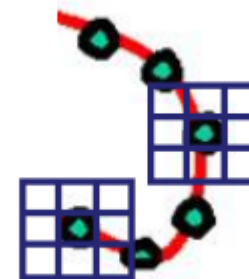
Representation of the contour

- Consider a discrete representation of the contour
- Consisting of a list of 2D points (vertices)

通过对一组控制点的移动，实现对轮廓的变化



At each iteration, we will have the option to move each vertex to another nearby location



Snake model 蛇形模型

Fitting deformable contours

How should we adjust the contour to form the new one at each iteration?

- Define a cost function (energy function) that indicates how good a candidate state is
- Seek next state that minimizes the cost function



High energy

Minimized energy

最优分割结果应当具有较低的能量

活动轮廓与边缘的吻合程度越高，能量越低

Snake model 蛇形模型



Energy function 能量函数

The total energy (cost) of the current snake is defined as:

$$E_{total} = E_{internal} + E_{external}$$

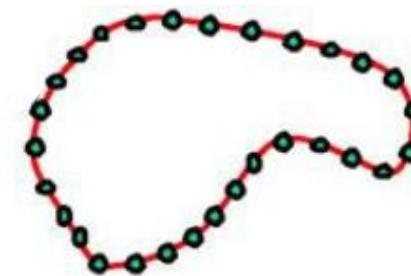
最优分割结果应当具有较低的能量

- Internal energy 内部能量

- Encourage prior shape preference
- E.g. smoothness, elasticity, particular known shape

- External energy (image energy) 外部能量

- Encourage contour to fit on places where image structure exists
- E.g., edges 活动轮廓与边缘的吻合程度越高，外部能量越低



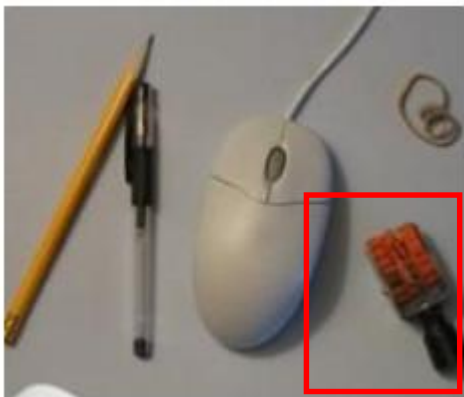
Snake model 蛇形模型

External energy 外部能量

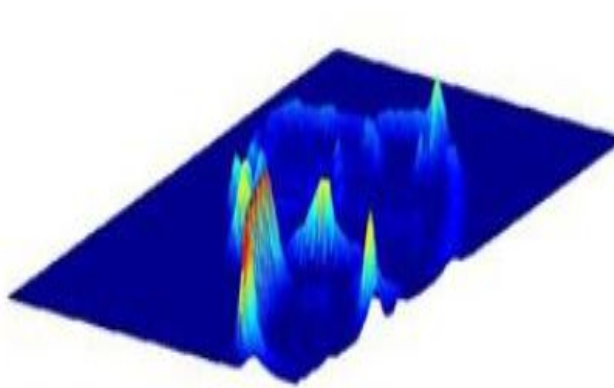
Intuition: we encourage the vertex to be on the edge in the image

One way is to define external energy as the **negative value of gradient magnitude**

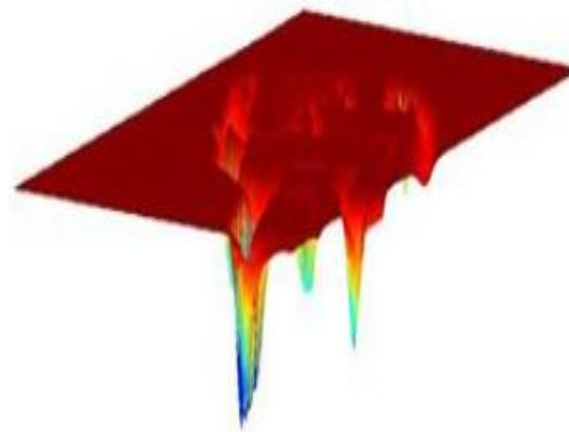
$$E_{external} = -(G_x(I)^2 + G_y(I)^2)$$



image



Magnitude of gradient
 $G_x(I)^2 + G_y(I)^2$



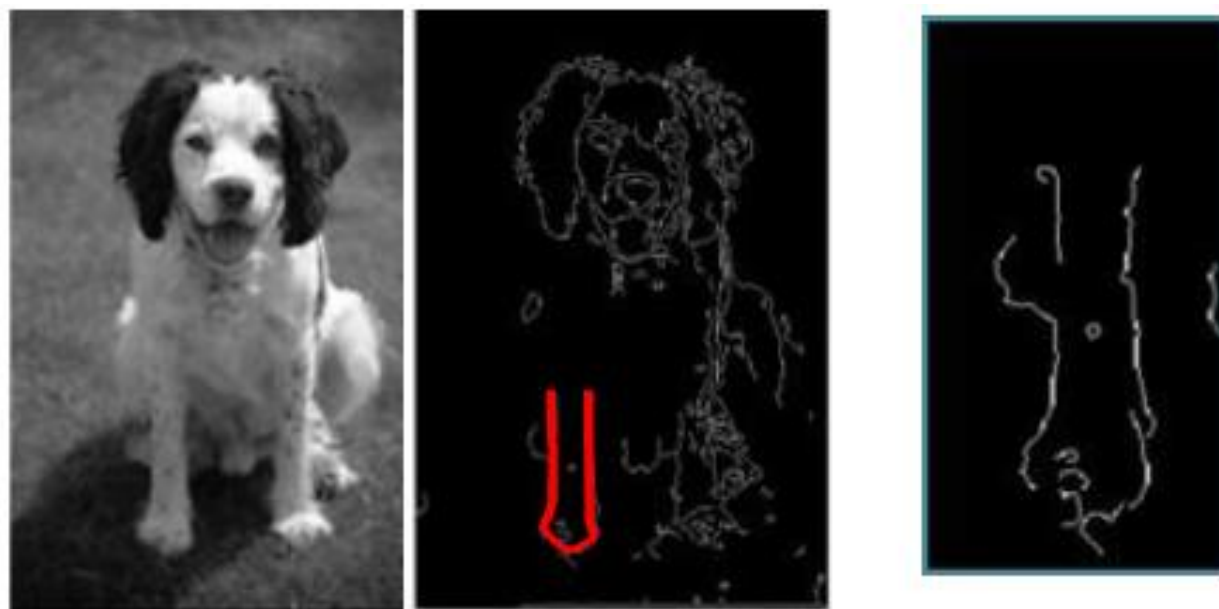
Magnitude of gradient
 $-(G_x(I)^2 + G_y(I)^2)$

梯度大的地方能量低

Snake model 蛇形模型

Internal energy 内部能量

We want to favor **smooth shapes**, contours with **low curvature**, contours similar to **a known shape**, etc. to balance what is actually observed (i.e., the gradient image)

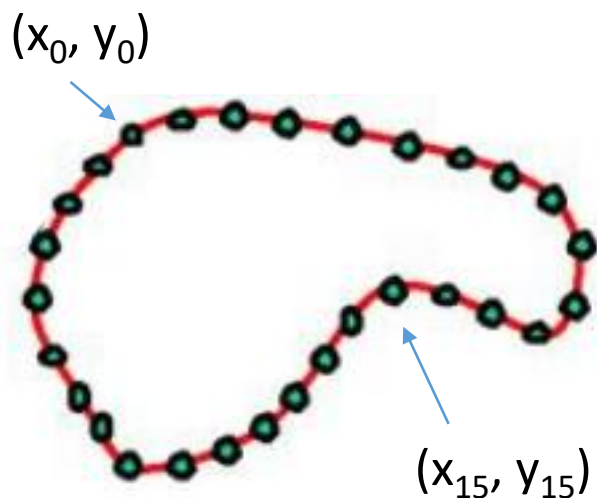


Snake model 蛇形模型

Internal energy 内部能量

We want to favor **smooth shapes**, contours with **low curvature**, contours similar to a **known shape**, etc. to balance what is actually observed (i.e., the gradient image)

A common internal energy is the “bending energy”



$$E_{internal}(v(s)) = \alpha \left| \frac{dv}{ds} \right|^2 + \beta \left| \frac{d^2v}{ds^2} \right|^2$$

Tension,
Elasticity
弹性约束

Stiffness,
Curvature
曲率约束



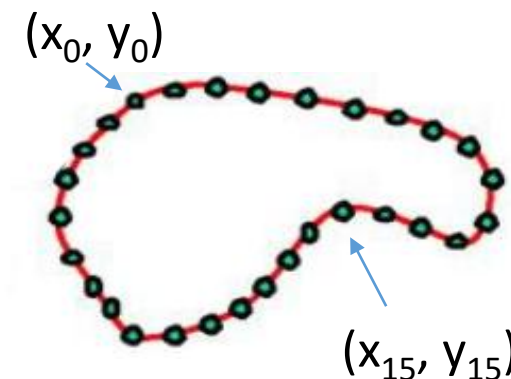
Elasticity 弹性（张力）

Snake model 蛇形模型

Internal energy

$$E_{internal}(v(s)) = \underbrace{\alpha \left| \frac{dv}{ds} \right|^2}_{\text{Tension, Elasticity}} + \underbrace{\beta \left| \frac{d^2v}{ds^2} \right|^2}_{\text{Stiffness, Curvature}}$$

$$\frac{dv}{ds} \approx v_{i+1} - v_i \quad \frac{d^2v}{ds^2} \approx (v_{i+1} - v_i) - (v_i - v_{i-1}) = v_{i+1} - 2v_i + v_{i-1}$$



Note these are **derivatives relative to position**, not spatial image gradient

$$E_{internal} = \sum_{i=0}^{n-1} \alpha \|v_{i+1} - v_i\|^2 + \beta \|v_{i+1} - 2v_i + v_{i-1}\|^2$$

弹性约束 曲率约束

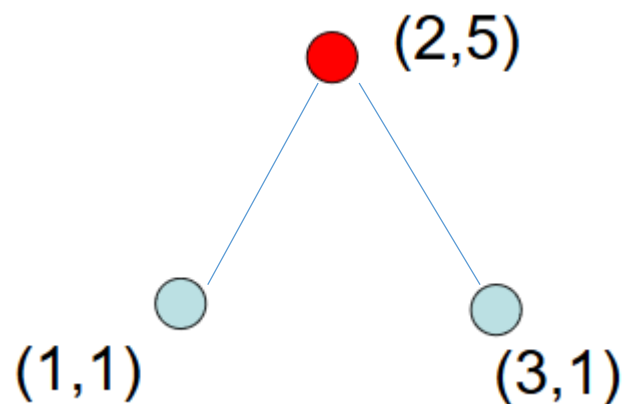
Snake model 蛇形模型



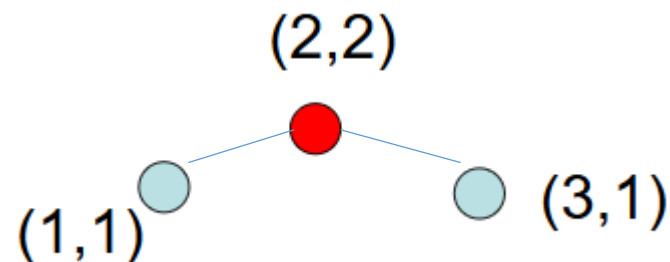
Internal energy

Understanding the curvature term

$$E_{curvature}(v_i) = \|v_{i+1} - 2v_i + v_{i-1}\|^2 \\ = (x_{i+1} - 2x_i + x_{i-1})^2 + (y_{i+1} - 2y_i + y_{i-1})^2$$



$$(3-2 \times 2+1)^2 + (1-2 \times 5+1)^2 \\ =64$$



$$(3-2 \times 2+1)^2 + (1-2 \times 2+1)^2 \\ =4$$

Snake model 蛇形模型

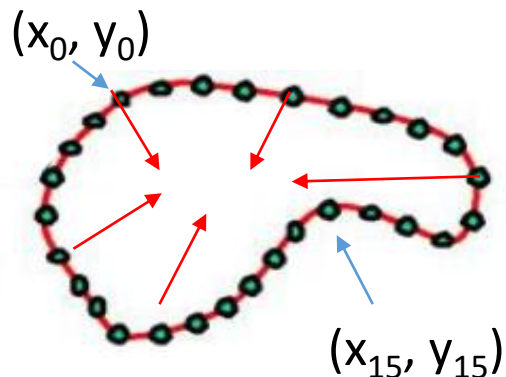
Penalizing elasticity

Current elastic energy uses a discrete estimate of the derivative

$$E_{elastic} = \sum_{i=0}^{n-1} \alpha \|v_{i+1} - v_i\|^2 = \alpha \cdot \sum_{i=0}^{n-1} (x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2$$

What is the possible problem with this definition?

The vertex tend to shrink to a single point in the center



Instead, we define elasticity as:

上一次迭代中的平均距离

$$E_{elastic} = \alpha \cdot \sum_{i=0}^{n-1} \left((x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 - \bar{d} \right)^2$$

Where d is the average distance between neighboring points, and it is updated at each iteration

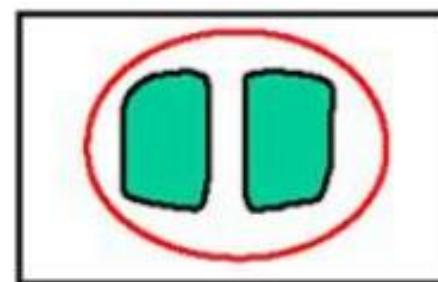
Snake model蛇形模型

Dealing with missing edges

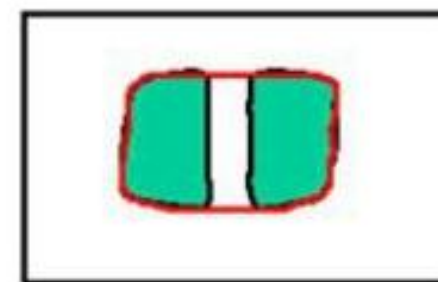
The preferences for low-curvature, smoothness help deal with missing edges



Aortic valve segmentation based on GVF-snake



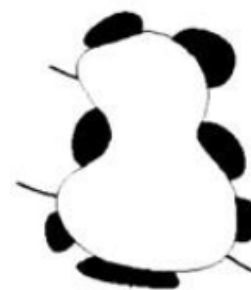
Initial contour



final contour



image



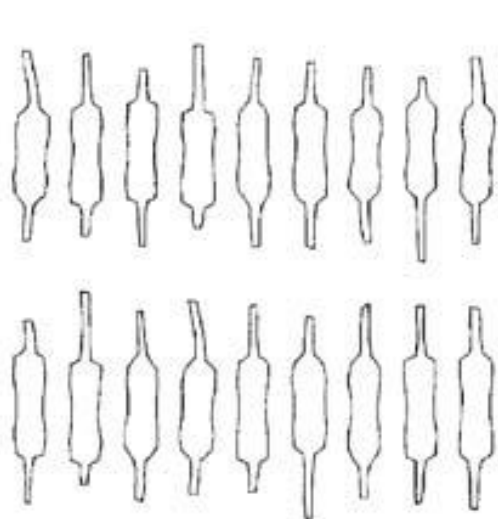
Segmentation contour

Snake model 蛇形模型

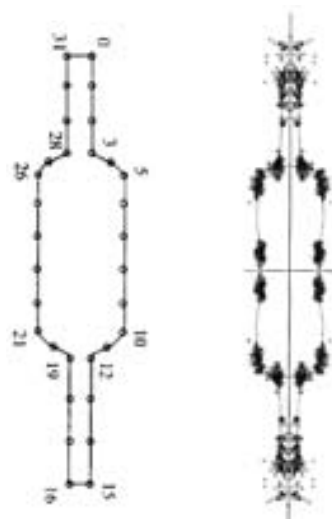
Using shape prior for snake 使用先验形状

Constraint the segmentation with a **statistical shape model** 统计形状模型

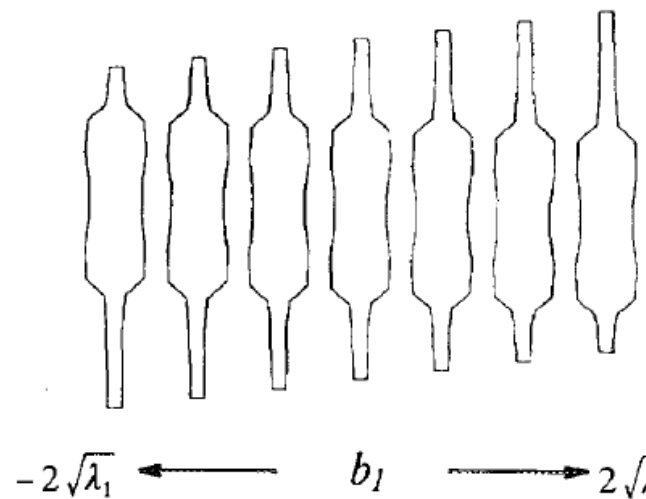
Build the shape model from a set of instances collected previously



A set of shapes



Control points and statistical model



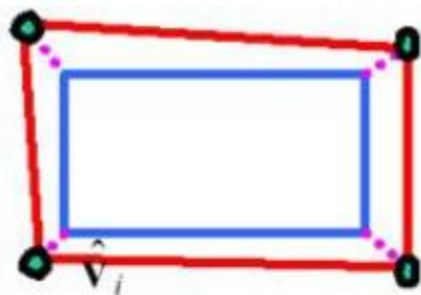
Shape controlled by the model parameter

从一组实例中构建统计形状模型

Snake model 蛇形模型

Using shape prior for snake 使用先验形状

Introduce an additional term to penalize deviation from that the shape model



$$E_{internal} += \alpha \cdot \sum_{i=0}^{n-1} (v_i - \hat{v}_i)^2$$

$\{\hat{v}_i\}$ are the points of the known shape

Snake model 蛇形模型



Advantages

Low complexity

Easy to introduce prior knowledge

Can account for open as well as closed structures

Disadvantages

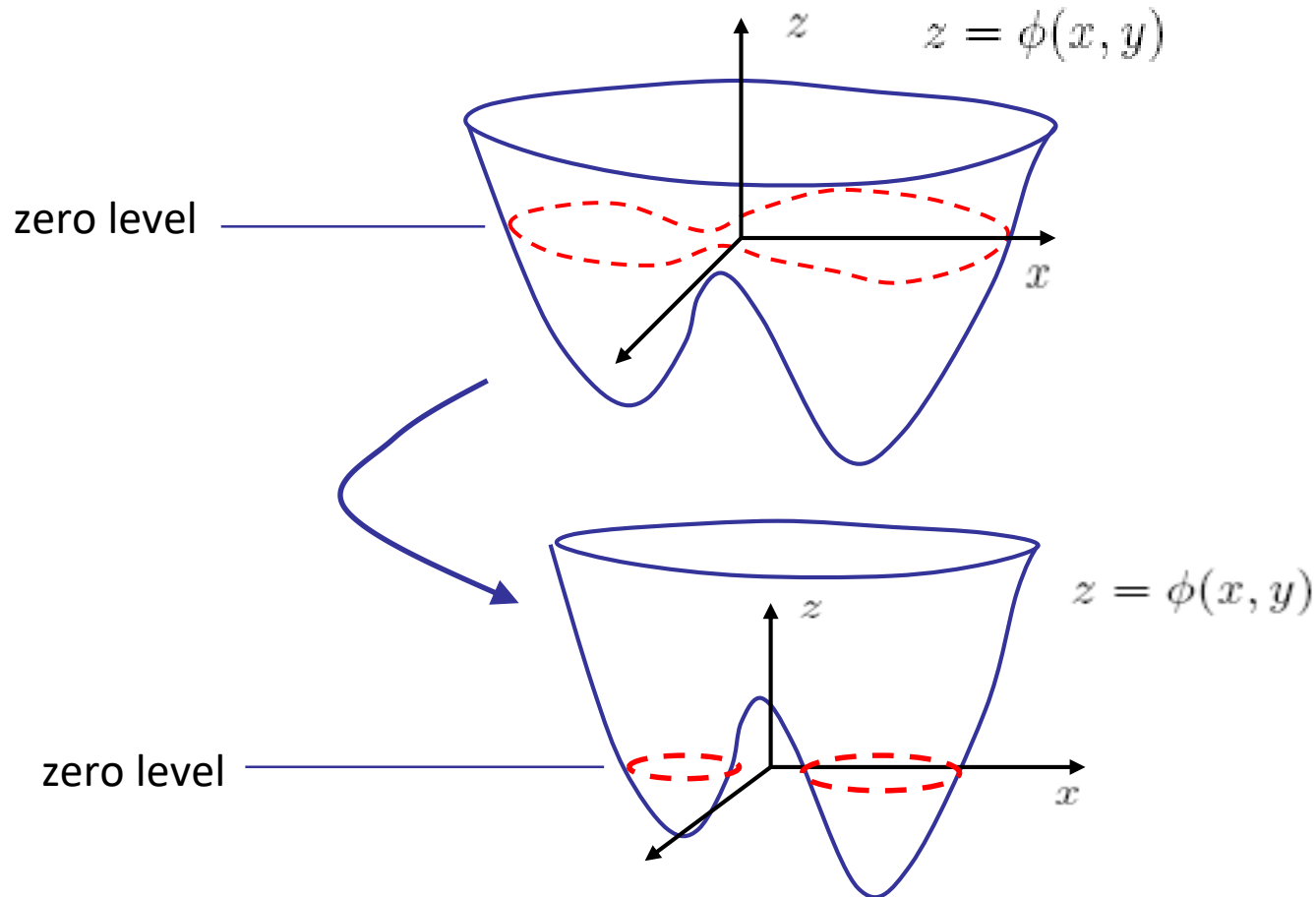
Sensitive to the initial contours 对初始轮廓敏感

Does not allow change of topology 不能改变形状的拓扑结构

Selection on the parameter space and the sampling rule affects the result

Level set 水平集

- Level set method represents the curve in the form of an **implicit surface**



Consider a function $\phi(x, y)$ is a surface

$$\phi(x, y) = z$$

The zero level set $\phi(x, y) = 0$ is a curve in the 2D space

Advantages

- All computation is carried out on a fixed grid, **no need for re-parameterization**
- Topological changes are handled naturally.**

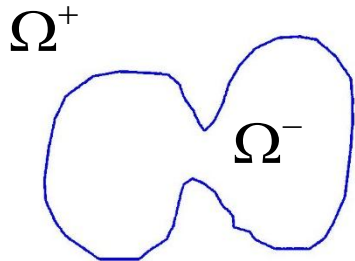
可改变形状的拓扑结构

Level set 水平集

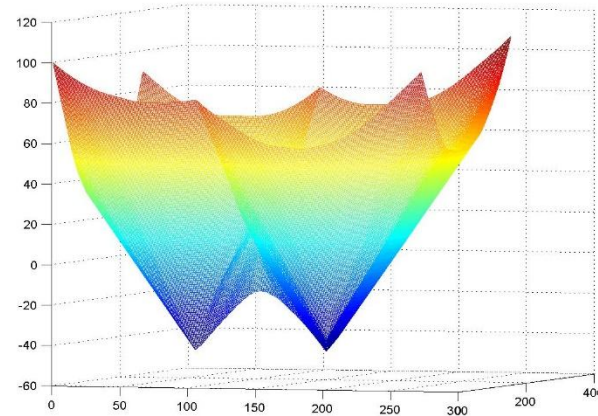
- Convert a contour into a level set function
- We can use signed distance function (SDF)

将曲线转换成曲面：距离变换

Contour C



Signed distance function



A signed distance function is defined by :

$$\phi(x) = \begin{cases} \text{dist}(x, C) & \text{if } x \text{ is outside } C \\ 0 & x \in C \\ -\text{dist}(x, C) & \text{if } x \text{ is inside } C \end{cases}$$

Level set 水平集



- Level set evolution

$$\frac{\partial \phi}{\partial t} = F |\nabla \phi|$$

F is a speed function, can be defined in different ways

- Constant speed
- Mean curvature flow
- Geodesic active contours

$$\frac{\partial \phi}{\partial t} = c |\nabla \phi|$$

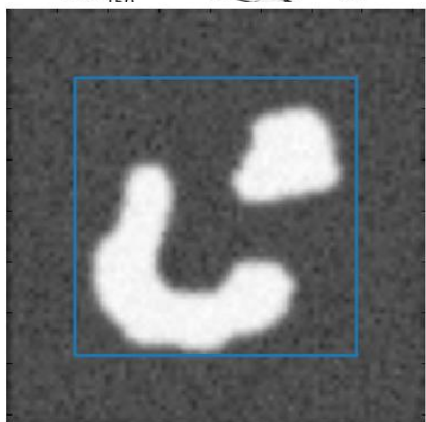
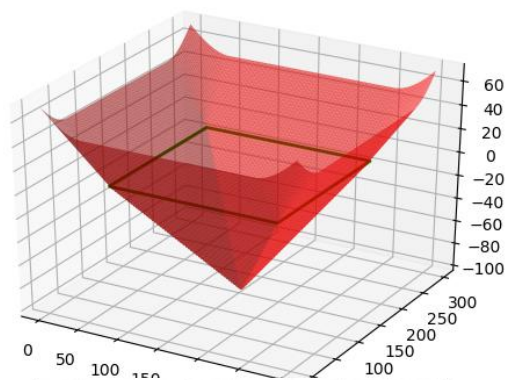
$$\frac{\partial \phi}{\partial t} = \operatorname{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) |\nabla \phi|$$

$$\frac{\partial \phi}{\partial t} = g(I) \|\nabla \phi\| \operatorname{div} \left(\frac{\nabla \phi}{\|\nabla \phi\|} \right) + g(I) \|\nabla \phi\| v + \nabla g(I) \cdot \nabla \phi$$

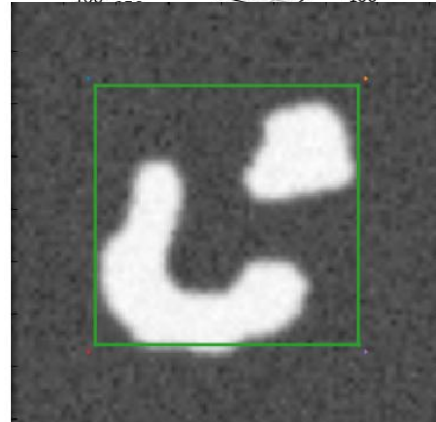
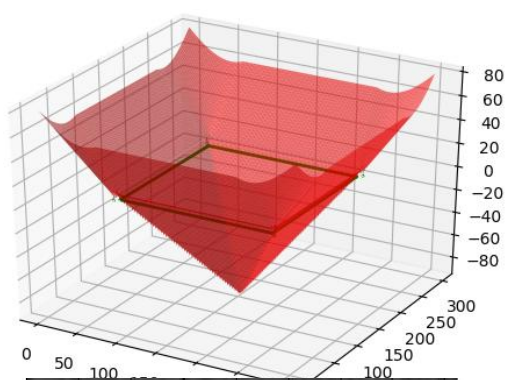
Level set 水平集

- Constant speed

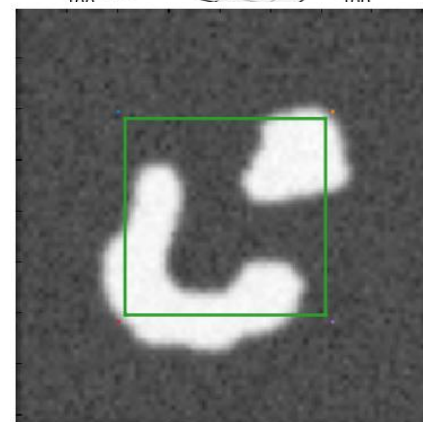
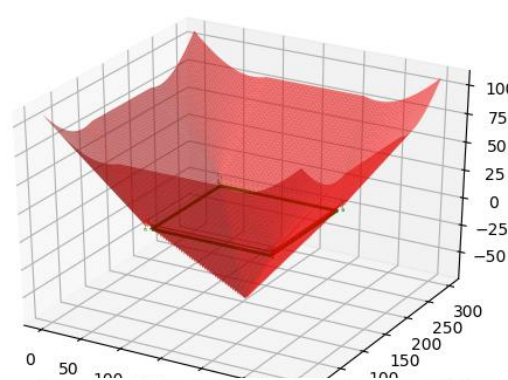
$$\frac{\partial \phi}{\partial t} = c |\nabla \phi|$$



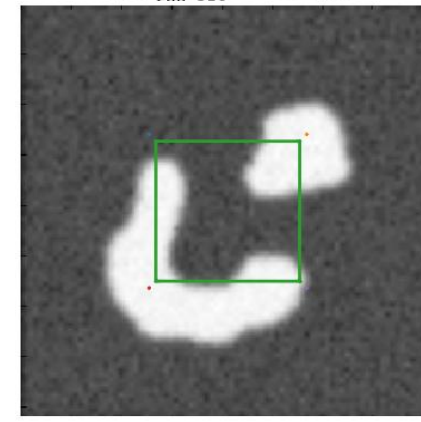
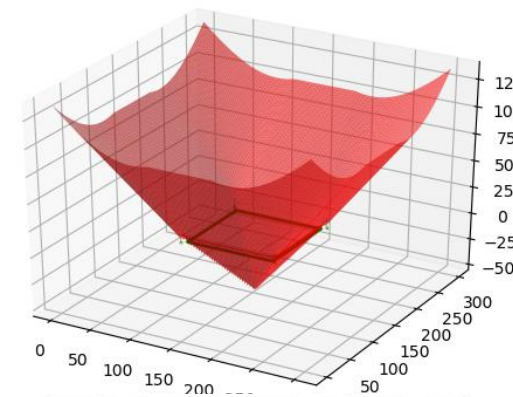
T = 0



T = 10



T = 30

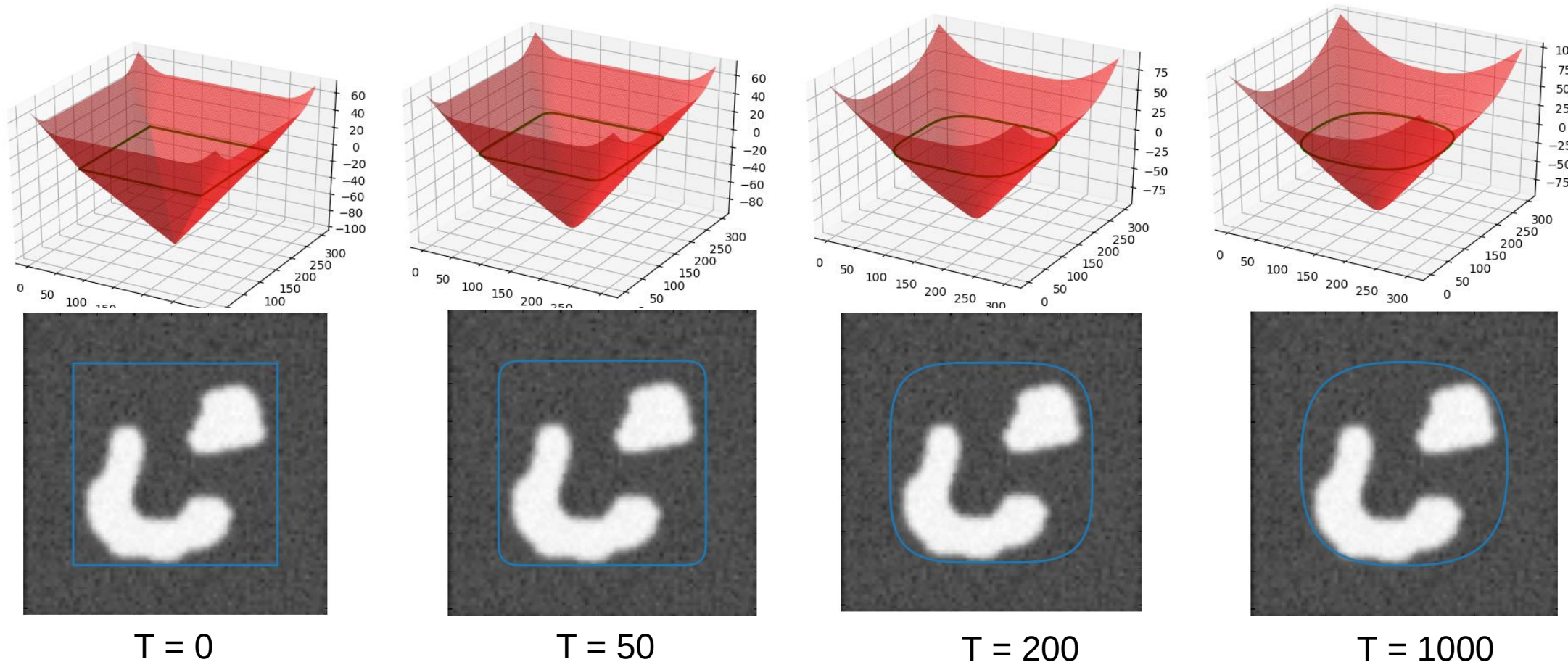


T = 50

Level set 水平集

- Mean curvature flow

$$\frac{\partial \phi}{\partial t} = \operatorname{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) |\nabla \phi|$$



Level set 水平集

- Define the speed based on gradient

$$\frac{\partial \phi}{\partial t} = F |\nabla \phi|$$

Stopping function

$$\text{Let } F = g(I) = \frac{1}{1 + \|\nabla I\|^2}$$



image



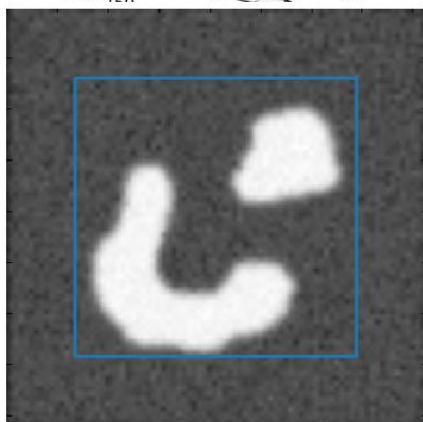
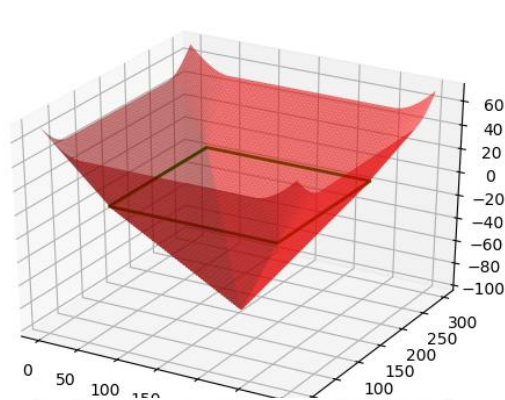
$g(I)$

- Large gradient leads to small $g(I)$ value
- Use $g(I)$ to define the speed
 - so that the level set stops at the edge

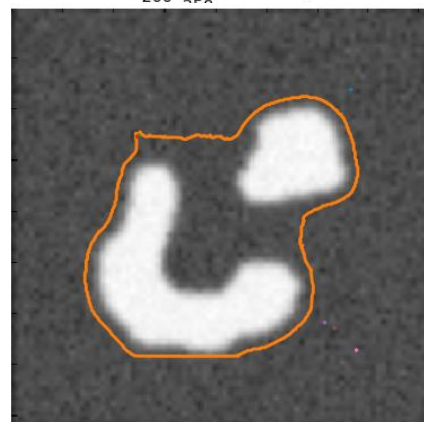
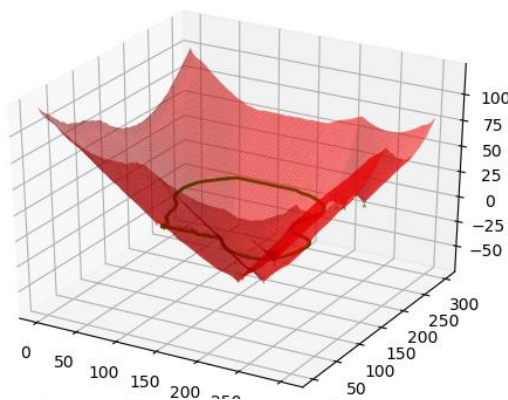
Level set 水平集

- Define the speed based on gradient $\frac{\partial \phi}{\partial t} = F|\nabla \phi|$

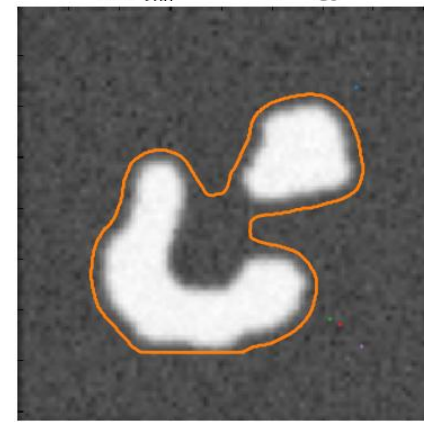
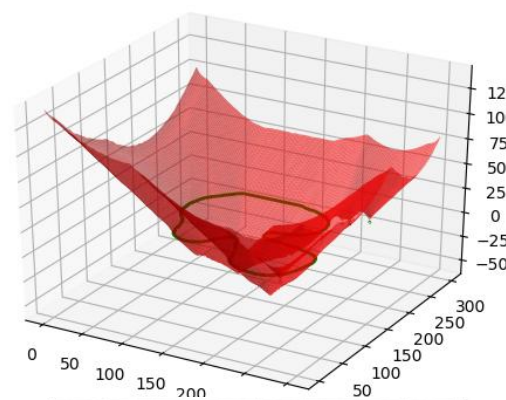
$$\text{Let } F = g(I) = \frac{1}{1 + \|\nabla I\|^2}$$



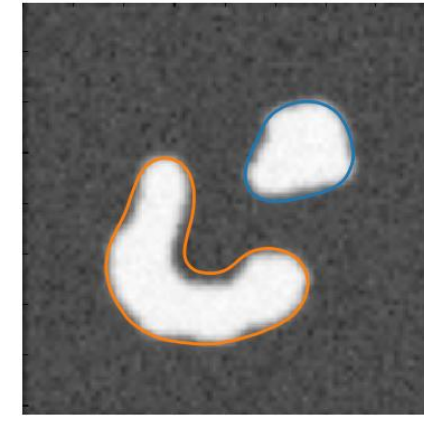
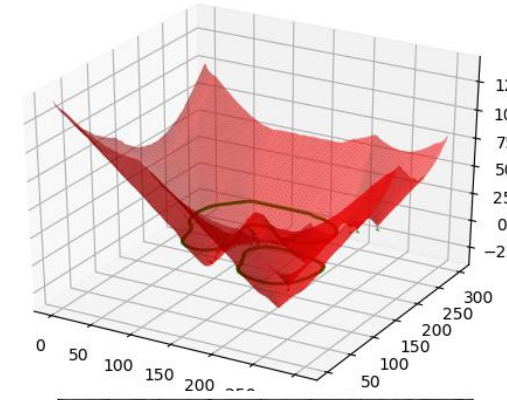
$T = 0$



$T = 50$



$T = 100$



$T = 200$

Level set 水平集



- Geodesic active contour

$$\frac{\partial \phi}{\partial t} = F |\nabla \phi|$$

$$\frac{\partial \phi}{\partial t} = g(I) \|\nabla \phi\| \operatorname{div} \left(\frac{\nabla \phi}{\|\nabla \phi\|} \right) + g(I) \|\nabla \phi\| v + \nabla g(I) \cdot \nabla \phi$$

Smooth term based
on curvature

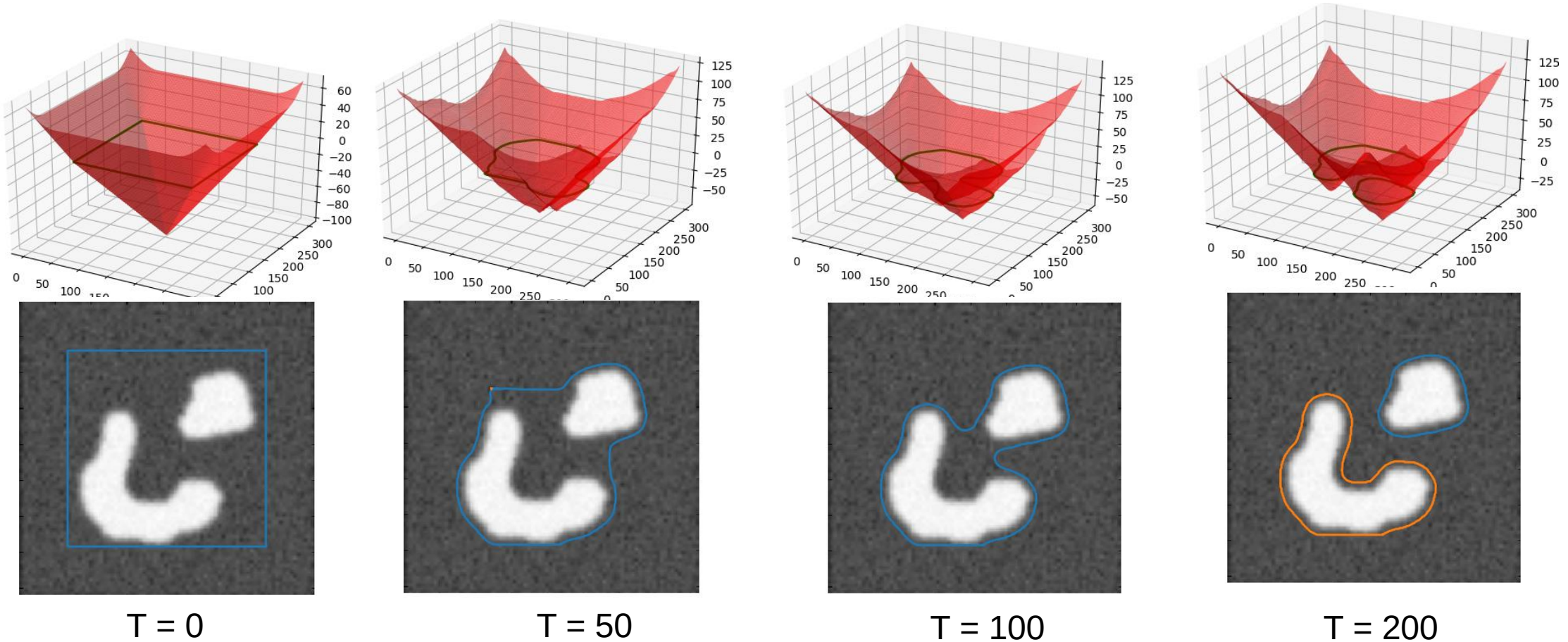
Balloon term,
controlling the speed of
the curve propagation
with parameter v .

attachment term, helps
the curve to converge

Level set 水平集

- Geodesic active contour

$$\frac{\partial \phi}{\partial t} = g(I) \|\nabla \phi\| \operatorname{div} \left(\frac{\nabla \phi}{\|\nabla \phi\|} \right) + g(I) \|\nabla \phi\| v + \nabla g(I) \cdot \nabla \phi$$



Region-based Level set 基于区域的水平集



Active contours without edges

TF Chan, LA Vese - IEEE Transactions on image processing, 2001 - ieeexplore.ieee.org

... edges too. In this paper, we propose a different active contour model, without a stopping edge-... In this way, we obtain a model which can detect contours both with or without gradient, for ...

☆ Save Cite Cited by 13614 Related articles All 50 versions

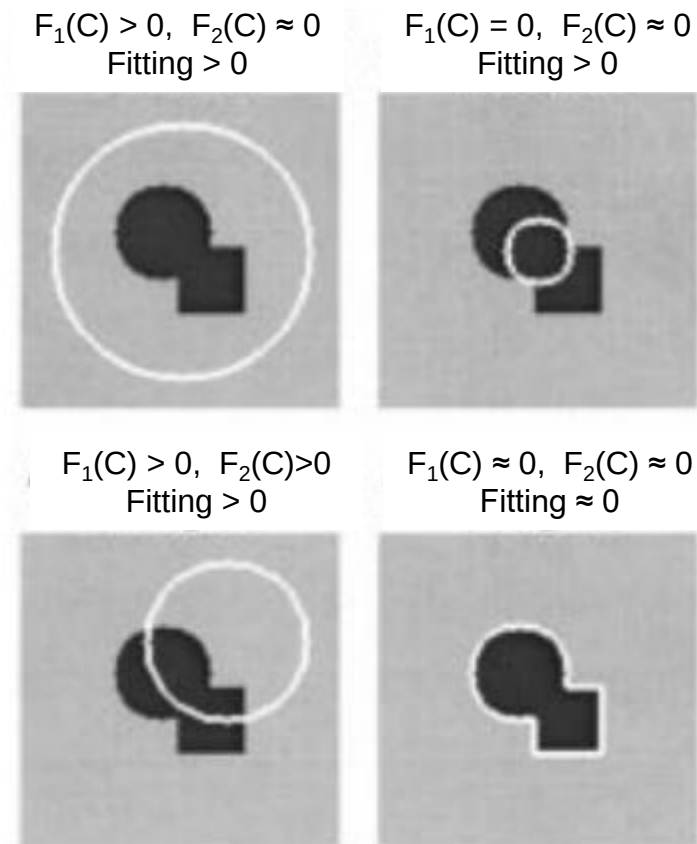
- Chan-Vese model: active contours without edges
- Limitation of edge-based level set
 - Cannot perform well if the edge is weak
 - The stopping function g is never zero on the edge
- Chan-Vese model defines a region-based energy function

$$g(I) = \frac{1}{1 + \|\nabla I\|^2}$$

类内方差最小化

$$F_1(C) + F_2(C) = \int_{\text{inside}(C)} |u_0(x, y) - c_1|^2 dx dy + \int_{\text{outside}(C)} |u_0(x, y) - c_2|^2 dx dy$$

C_1 and C_2 are average value of inside and outside regions, respectively.



Region-based Level set 基于区域的水平集



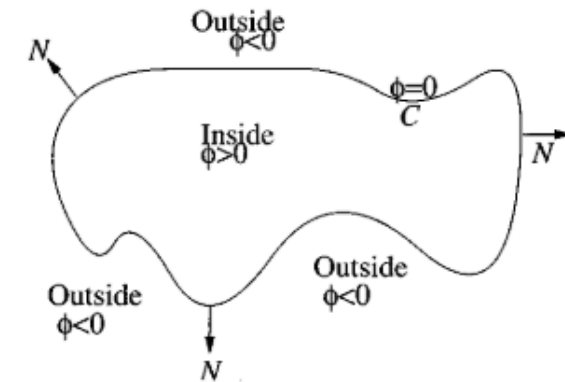
- Chan-Vese model: active contours without edges
- To add some regularizations, constraint the length and area of the curve

轮廓长度 区域面积

$$F(c_1, c_2, C) = \mu \cdot \text{Length}(C) + \nu \cdot \text{Area}(\text{inside}(C)) \\ + \lambda_1 \int_{\text{inside}(C)} |u_0(x, y) - c_1|^2 dx dy \\ + \lambda_2 \int_{\text{outside}(C)} |u_0(x, y) - c_2|^2 dx dy$$

Convert curve
representation
into level set
representation

$$F_\varepsilon(c_1, c_2, \phi) \\ = \mu \int_{\Omega} \delta_\varepsilon(\phi(x, y)) |\nabla \phi(x, y)| dx dy \quad \text{Length term} \\ + \nu \int_{\Omega} H_\varepsilon(\phi(x, y)) dx dy \quad \text{Area term} \\ + \lambda_1 \int_{\Omega} |u_0(x, y) - c_1|^2 H_\varepsilon(\phi(x, y)) dx dy \quad \text{Inside region variance} \\ + \lambda_2 \int_{\Omega} |u_0(x, y) - c_2|^2 (1 - H_\varepsilon(\phi(x, y))) dx dy \quad \text{Outside region variance}$$

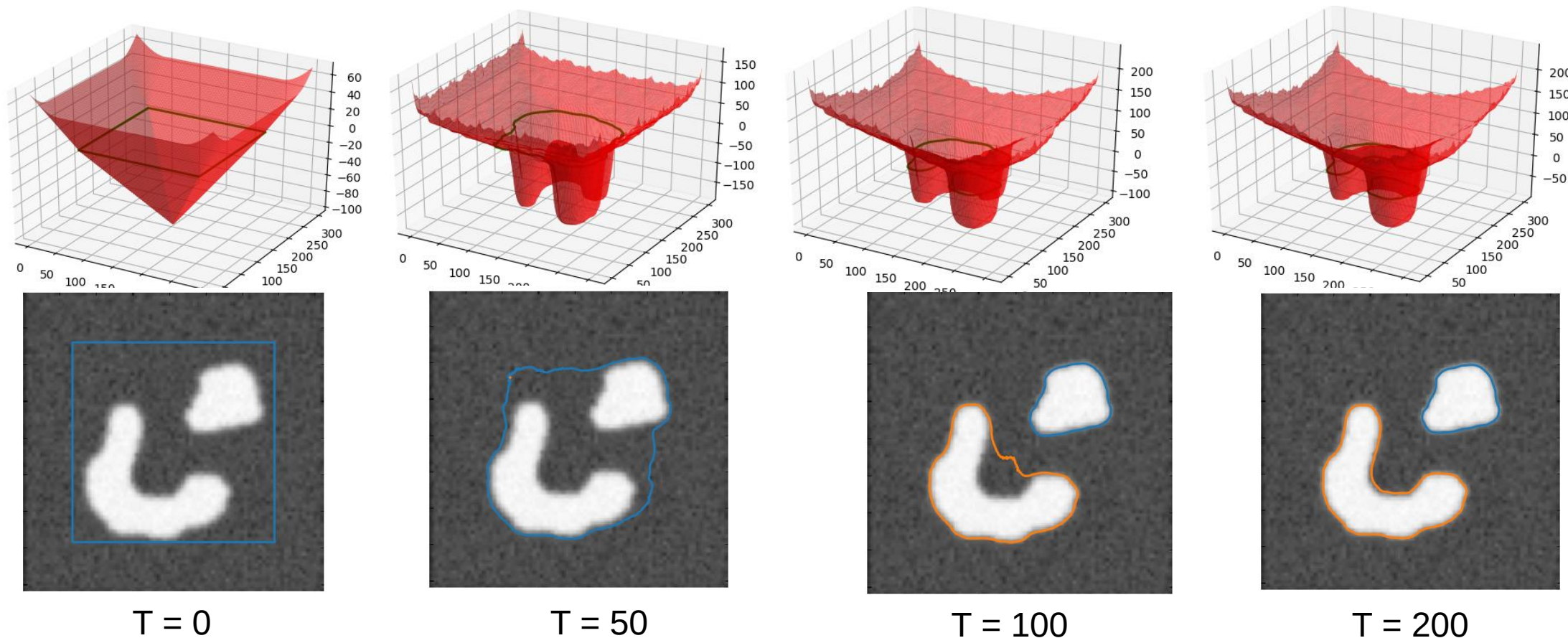


δ_ε : 脉冲函数

H_ε : 阶跃函数

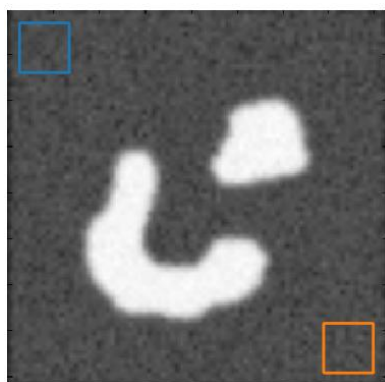
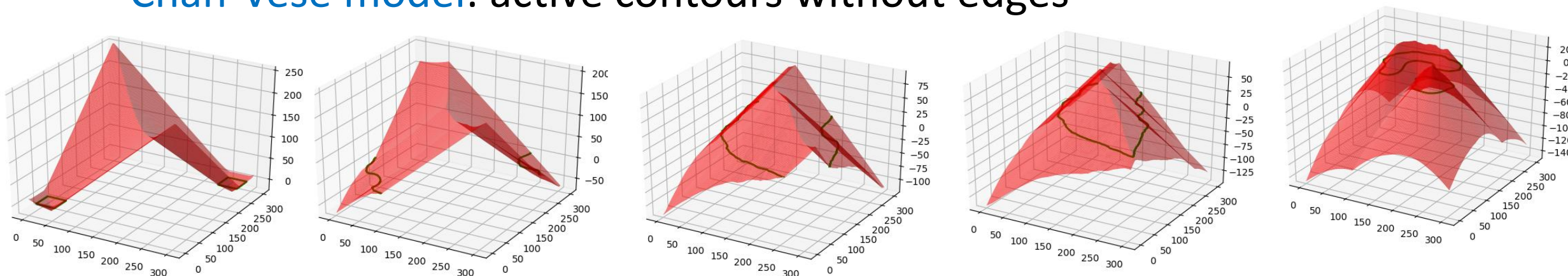
Region-based Level set 基于区域的水平集

- Chan-Vese model: active contours without edges

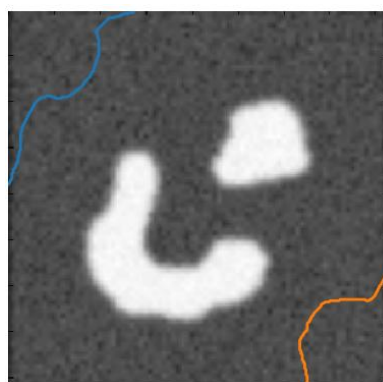


Region-based Level set 基于区域的水平集

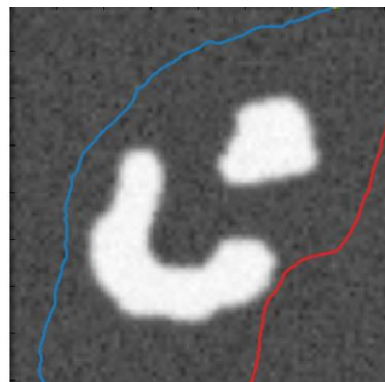
- Chan-Vese model: active contours without edges



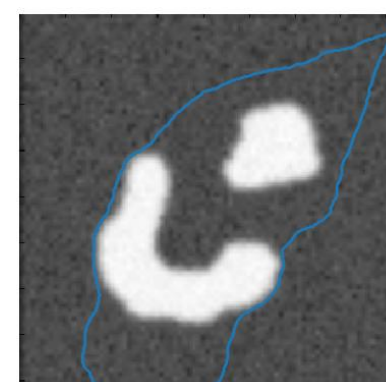
$T = 0$



$T = 200$



$T = 500$



$T = 600$

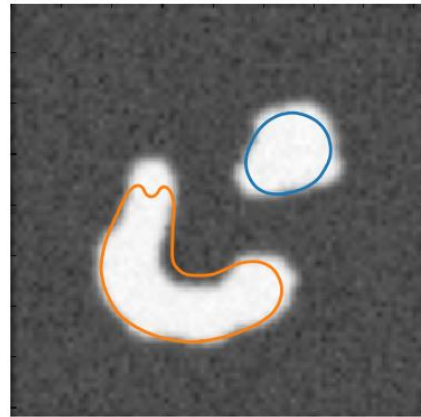
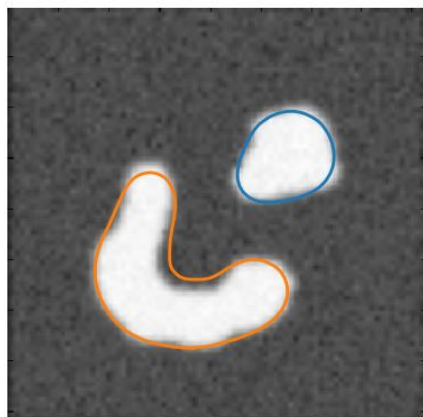
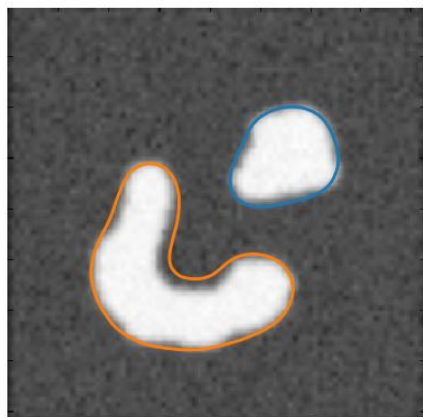


$T = 800$

Region-based Level set 基于区域的水平集



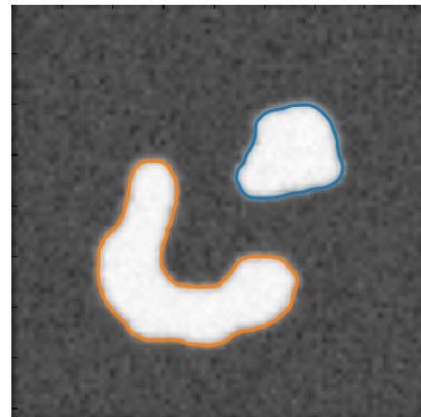
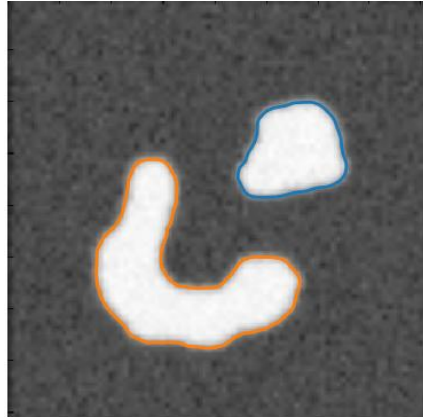
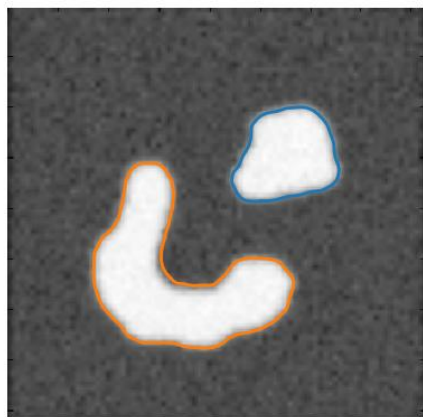
- Comparison between edge-based and Chan-Vese model



Edge-based level set

May pass through the boundary due to non-zero stopping function

$$g(I) = \frac{1}{1 + \|\nabla I\|^2}$$



Chan-Vese model

No need edge detection
Better fit the boundary

T = 200

T = 300

T = 400

小结：图像分割的基本方法

1, Intensity-based Segmentation

Thresholding 阈值法

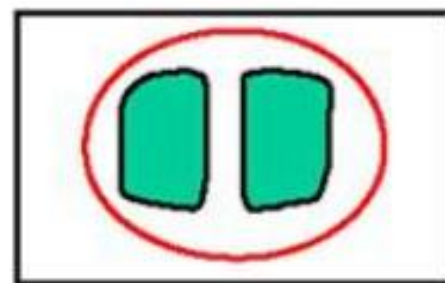
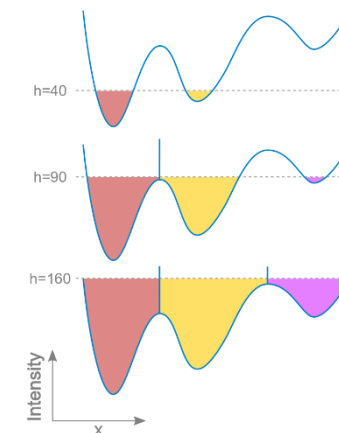
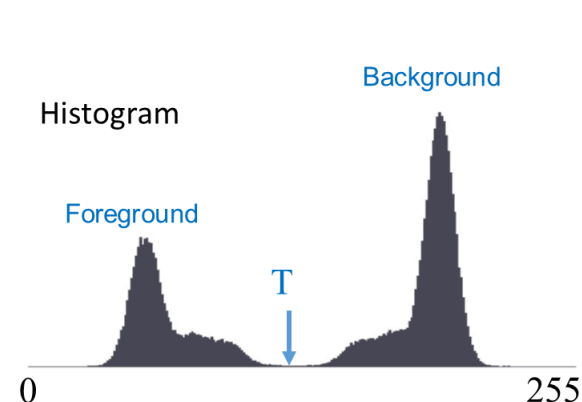
Watershed 分水岭算法

2, Clustering-based Segmentation

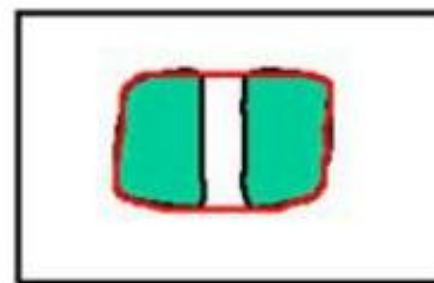
3, Active contours

Snake 蛇形模型

Level set 水平集



Initial contour



final contour

